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Paleomagnetism and geochronology of mafic dykes from the Southern Granulite Terrane, India: Expanding the Dharwar craton southward

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ABSTRACT

The Southern Granulite Terrane (SGT) of India is a collage of Archean to Neoproterozoic crustal blocks hosting several generations of mafic dykes. We report new paleomagnetic and geochronologic results from mafic dykes in the Salem Block of the SGT. The 2370 Ma Bangalore dyke swarm is confirmed in this area with a U-Pb zircon age of 2363 ± 6.6 Ma on dykes with steep dual-polarity paleomagnetic directions yielding a pole at 2° N and 73° E (A95 = 14°). These results precisely place the Dharwar Craton and northern SGT at high latitudes in the early Paleoproterozoic. The episodic Paleoproterozoic magmatic record of India is also recovered from the SGT. Several groupings of SGT paleomagnetic directions compare favorably with the 2.21 Ga, 2.18 Ga, and 1.88 Ga Dharwar craton dykes. We also report a grand mean pole of 81° N and 259° E (A95 = 11°) that we attribute to a widespread Ediacaran-age remagnetization associated with the final stages of Gondwana assembly. These paleomagnetic and geochronologic data permit a better resolved reconstruction of Indian crustal elements from the Paleoproterozoic onward.

1. Introduction

1.1. Rationale

Crustal elements of the Indian subcontinent have a place in every supercontinent reconstruction, from Ur to Pangea (Rogers and Santosh, 2003; Meert, 2012; Meert, 2014). Suitable target rocks for paleomagnetic study throughout peninsular India are available to aid in the reconstruction of India within the Proterozoic supercontinents of Kenorland, Columbia (Nuna) and Rodinia. The relationship of major tectonic elements of the Indian shield has been a matter of debate *vis-a-vis* the timing of final assembly of the five major Indian cratons (Aravalli, Bundelkhand, Singhbhum, Bastar, and Dharwar). The ENE-WSW trending Central Indian Tectonic Zone (CITZ) separates the North Indian Block (NIB comprised of Aravalli and Bundelkhand cratons) and South Indian Block (SIB - Singhbhum, Bastar, and Dharwar cratons). Proterozoic tectonothermal events in the CITZ can be resolved into three pulses of tectonothermal activity at approximately 2.5 Ga (Stein et al., 2004, 2014), between 1.8 and 1.5 Ga and again from 1.0–0.9 Ga (Bhandari et al., 2011; Bhowmik et al., 2012a; Bhowmik et al., 2012b; Meert et al., 2010, 2011; Meert and Pandit, 2015). As even a brief review of the conflicting arguments on the CITZ is beyond the scope of this paper, we merely note that deep-seated disagreement exists regarding the timing of final assembly of the NIB and the SIB. This is precisely where paleomagnetism is useful as a test of the null-hypothesis. While the *lack* of concordant, contemporaneous paleomagnetic data linking the various pieces of the puzzle do not intrinsically discount their mutual linkages, discordant, contemporaneous paleomagnetic data certainly do. Alternatively, and more positively, coeval and concordant paleomagnetic poles between different blocks provide strong evidence for their juxtaposition. Fortunately, the Precambrian shield regions of peninsular India are cross-cut by extensive Proterozoic-age mafic dyke swarms suitable for paleomagnetic studies.

The Southern Granulite Terrane (SGT) located to the south of the Dharwar Craton and north of the Palghat-Cauvery Shear Zone (Fig. 1), hosts numerous phases of mafic dykes; however, paleomagnetic data from those dykes are less well-resolved than elsewhere in India (Dash et al., 2013; Radhakrishna et al., 2013b). In part, this is due to the fact that some of the dykes show clear evidence of alteration. In keeping with our “India first” approach to Precambrian paleogeographic reconstructions, we examine connections between the Salem (Northern) Block of the SGT with the Dharwar Craton to the north. Current consensus, based on isotopic and geochemical studies, is that the Salem Block and other northern granulites are the metamorphosed equivalents

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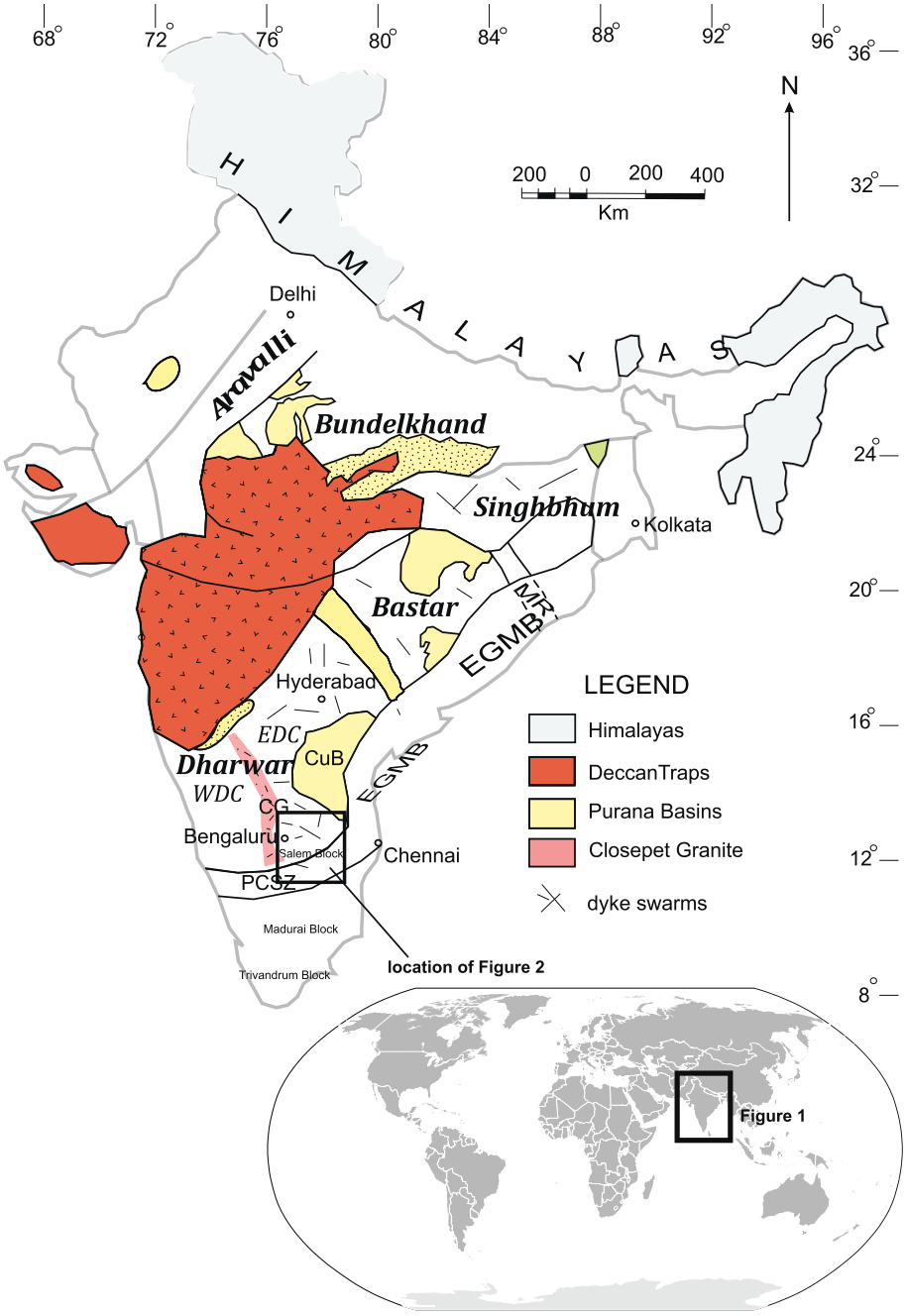


Fig. 1. Map of India showing major cratonic nuclei and Proterozoic-aged sedimentary basin features (adapted from Meert and Pandit, 2015). Box indicates Fig. 2 location.

of the Dharwar Craton (Collins et al., 2014). Recent paleomagnetic work supports this contention, as a steep paleomagnetic direction from mafic dykes within the SGT was dated to 2318 ± 60 Ma with a Sm-Nd whole-rock isochron age (Dash et al., 2013) that agrees, within error, with the 2370 ± 1 Ma U-Pb baddeleyite age from the "Bangalore swarm" of dykes in the Dharwar Craton (Halls et al., 2007). We have extensively sampled mafic dykes from the northern SGT to test their correlation with the greater Dharwar craton and report these new paleomagnetic and geochronological data in this paper.

1.2. Regional geologic setting and area of study

The geological history of Peninsular India revolves around five major cratonic nuclei – the Aravalli, Bundelkhand, Singhbhum, Bastar, and Dharwar cratons (Fig. 1). A collage of crustal blocks of various ages collectively make up the Southern Granulite Terrane (SGT) to the south of Dharwar Craton (Meert et al., 2010, 2011, Santosh et al., 2017). The older northern blocks of the SGT are separated from the southern (Neoproterozoic) blocks by the Palghat-Cauvery Shear Zone (PCSZ). This shear zone is thought to represent an Ediacaran-Cambrian suture formed during the final assembly of Gondwana (Collins et al., 2007).

Southern India can be divided into two main geologic terranes: the Dharwar Craton (DC) in the north and the Southern Granulite Terrane in the south (see Fig. 1). The Dharwar Craton is further subdivided into Eastern Dharwar Domain (EDD) and Western Dharwar Domain (WDD) separated by the Closepet granite (Ramakrishnan and Vaidyanadhan, 2008; Meert and Pandit, 2015). Large intracratonic basins are present throughout the Dharwar Craton, typified by the massive, crescent-shaped Cuddapah Basin in the EDD. The metamorphic grade increases southward in the Dharwar Craton and reaches its highest grade in the Southern Granulite Terrane south of the Fermor Line (i.e. the charnockite/non-charnockite boundary; Fermor, 1936).

The SGT consists of three late Archaean to Neoproterozoic age, high-grade metamorphic blocks (Fig. 1); including the Salem (Northern) Block, the Madurai Block, and the Trivandrum Block (Drury et al., 1984). The SGT crustal blocks are separated by several large-scale shear zones (Meert et al., 2010, 2011; Collins et al., 2014; Pandit et al., 2016). The northernmost unit of the SGT, the Salem Block, is bounded to the south and east by the Moyar–Bhavani and Attur Shear Zones (MBSZ and ASZ), respectively. The Madurai and Trivandrum blocks occur south of the Palghat-Cauvery Shear Zone. The main lithologies present in the Salem Block are orthogneisses and corresponding charnockites, with subordinate mafic granulites and metasedimentary rocks (Collins et al., 2014). Major metamorphic intervals range in age from the late Archean to the earliest Proterozoic (Peucat et al., 1995, Ghosh et al., 2004; Clark et al., 2009, Anderson et al., 2012; Collins et al., 2014). Charnockitic rocks in the Salem block have been dated to 2538 ± 6 Ma and 2529 ± 7 Ma via SHRIMP Pb-Pb dating (Clark et al., 2009). Distinct, younger zircon rim ages of 2473 ± 8 Ma and 2482 ± 15 Ma were interpreted by Clark et al. (2009) to reflect timing of accretion of the Salem block to the Dharwar Craton. Neoproterozoic metamorphic U-Pb zircon and monazite ages of ca. 880 Ma and 730 Ma (Ghosh et al., 2004; Bhutani et al., 2007) indicate tectonothermal activity at these times, although causes are “cryptic” (Collins et al., 2014). Later metamorphism occurred during the Cambrian (Ghosh et al., 2004); however, the spatiotemporal distribution of this metamorphism is poorly constrained (Collins et al., 2014).

The Salem Block, like the adjoining Dharwar Craton to the north, is infested by a number of mafic dyke swarms. The dykes are concentrated in the Krishnagiri-Hosur area, near Dharmapuri, and around Tiruvannamalai (Fig. 2). Trends of these dykes vary but the majority trend E-W to NW-SE. The K-Ar ages for the Dharmapuri and Tiruvannamalai “swarms” range from ~2300 to 1350 Ma and ~2300 to 1650 Ma, respectively (Radhakrishna et al., 1986, 1999). A Sm-Nd whole-rock age from a dyke near Tiruvannamalai yielded an age of 2318 ± 60 Ma (Dash et al., 2013; Fig. 2). Dykes sampled in this study intrude both granitic gneiss and charnockite hosts. Some dykes in the northern regions of the SGT can be traced over long distances in the field or in satellite images. The dykes range in size from a few meters to tens of meters and often show sharp contacts with the host rocks.

1.3. Previous paleomagnetic work

Paleomagnetic data from mafic dykes in the SGT are limited (Venkatesh et al., 1987; Radhakrishna and Joseph, 1996; Dash et al., 2013). The pioneering paleomagnetic study of SGT mafic dykes sampled dykes from Kunnam, Tiruvannamalai and Attur (Venkatesh et al., 1987). In that study, remanences were subdivided into four groups with the following declination (D) and inclination (I) values: (a) D = 86°, $I = -35°$ ($\alpha_{95} = 5°$), (b) D = 33°, $I = -48°$ ($\alpha_{95} = 18°$), (c) D = 133°, $I = -75°$ ($\alpha_{95} = 6°$), and (d) D = 294°, $I = -52°$ ($\alpha_{95} = 23°$). The (c) group results are in general agreement with the well-established steeply inclined remanence of the giant “Bangalore swarm”, dated at 2370 Ma by Halls et al. (2007). This steeply inclined direction is ubiquitous throughout the Dharwar Craton (Radhakrishna et al., 2013; Belica et al., 2014). The (d) group of directions is similar to those found in the 2.21 Ga dykes from the Dharwar Craton (Kumar et al., 2012a).

Radhakrishna and Joseph (1996) argued for multiple dyke swarms in the Dharmapuri and Tiruvannamalai regions of the SGT. The analytical process followed by Radhakrishna and Joseph (1996) was to divide the dykes by arbitrary geographical regions without regard to other field information (trends/composition). That approach is questionable given that a dyke from Dharmapuri (D6) which exhibited a paleomagnetic signature characteristic of most dykes from Tiruvannamalai (steep negative inclination) was discarded under this scheme. Dyke trends noted in their paper were also relatively uniform for each area, which contrasts with our observations from the same regions. Their restrictive analysis resulted in 4 ‘distinct’ groupings which they named the Agali-Anaikatti dykes (D = 13.9°, I = +27°, $\alpha_{95} = 26°$, based on 2/9 sites); the Dharmapuri dykes (D = 360°, I = +9°, $\alpha_{95} = 11°$, based on 5/10 sites); the Tiruvannamalai dykes (D = 107°, $I = -78°$, $\alpha_{95} = 9°$, based on 7/14 sites); and the North Kerala dykes (D = 217°, $I = -50°$, $\alpha_{95} = 25°$, based on 2/5 sites).

Dash et al. (2013) focused on two concentrations of mafic dykes in the Southern Granulite Terrane. These are known as the East Coast Dykes near Chennai and the Tiruvannamalai “swarm”. The latter area has been sampled in four separate paleomagnetic investigations – including three studies with radiometric ages (Radhakrishna and Joseph, 1996; Dash et al., 2013; this study). Dash et al. (2013) obtained a direction from NW-SE trending Tiruvannamalai dykes and combined them with data from group (c) of Venkatesh et al. (1987) and Radhakrishna and Joseph (1996) to obtain a group mean (15 dykes) of D = 125°, $I = -74°$ ($\alpha_{95} = 8°$) for the Tiruvannamalai dykes. Dash et al. (2013) report a whole-rock Sm-Nd isochron age of 2318 ± 60 Ma which they correlated with the 2367 Ma “Bangalore” swarm of Halls et al. (2007) in contrast to the previously reported K-Ar age of 1650 ± 10 Ma (Radhakrishna et al., 1999). The primary nature of the steep paleomagnetic direction in Tiruvannamalai dykes was confirmed by a positive baked contact test (Dash et al., 2013). The East Coast dykes yielded a group mean direction (10 sites, 5 dykes) of D = 89°, I = +34° ($\alpha_{95} = 7°$) from both NW-SE and E-W trending dykes combined with group (a) of Venkatesh et al. (1987) and some “anomalous” directions of Radhakrishna and Joseph (1996).

2. Methods

2.1. Paleomagnetic analysis

A total of 51 sites were collected from dykes throughout the Southern Dharwar and northern Southern Granulite Terrane, primarily from quarries, roadcuts or areas where there was no ambiguity about the in-situ nature of the outcrop (Fig. 2). Samples were also collected from baked and unbaked host rocks at sites suitable for baked contact tests. Paleomagnetic samples were collected in the field with a water-cooled, gasoline-powered drill and oriented with magnetic and sun compass with the exception of three sites where drilling was not feasible. All samples were returned to the University of Florida where they were trimmed to a standard size. Natural remanent magnetization (NRM) directions were measured using either a Molspin spinner magnetometer or 2G-77R cryogenic magnetometer. Pilot samples from all sites were demagnetized by thermal and alternating methods using either an ASC TD-48 thermal demagnetizer or DTech 2000 AF demagnetizer. Subsequent demagnetization treatment was optimized based on the results from the pilot samples. Paleomagnetic vector directions were determined via principal component analysis (Kirschvink, 1980) using IADP software (Torsvik et al., 2016). Powdered material from several samples at each site was analyzed with a KLY-3S Kappabridge with a CS-3 furnace attachment and a vibrating sample magnetometer (VSM) in order to ascertain magnetic carriers and magnetic domain characteristics.

Approximately 2–3 kg of samples were collected from the coarse-grained centers of two dykes for geochronological investigation (sites I1510 and I1535). Standard mineral separation techniques (crushing, sieving, heavy liquids and magnetic methods) were used to isolate zircons from the dyke samples.

2.2. U–Pb geochronology

Zircon grains were hand-picked under a binocular microscope then transferred to double-sided tape in preparation for mounting. Site I1510 (SW Tiruvannamalai) yielded thirteen grains suitable for analysis. Site I1535 (near Krishnagiri) yielded 8 zircons for analysis. The grains were set in epoxy then polished to expose interiors. Zircons were examined on an SEM with backscatter electron (BSE) and cathode luminescence

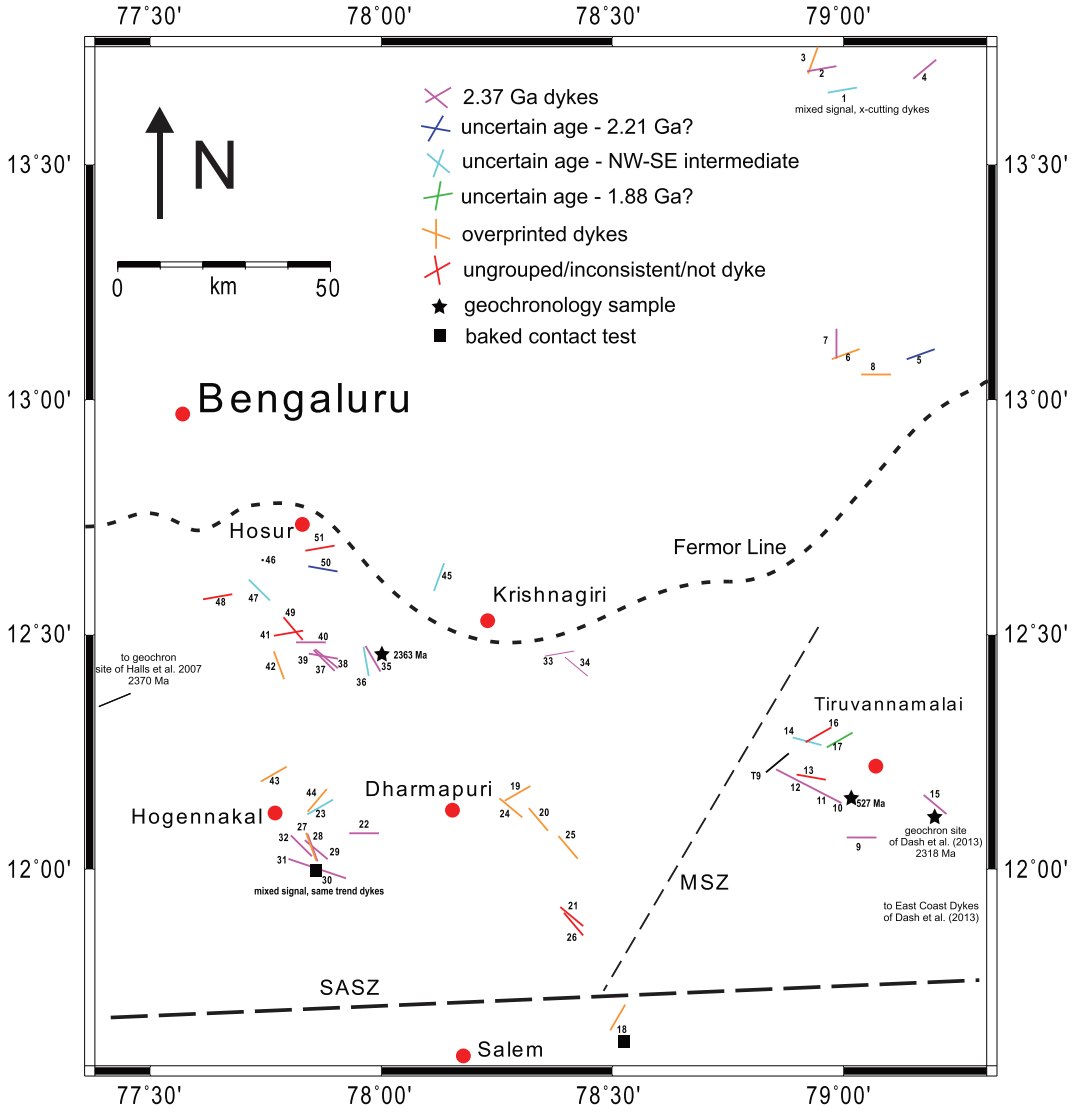


Fig. 2. Regional map of dykes sampled in this study. Groups of directions are represented by different colored line segments. Dyke trends shown were determined from field observations and satellite imagery. Sites of baked contact tests and geochronological sampling are shown.

Table 1
Geochronological results for site 10.

Grain name	$^{207}Pb/^{235}U$	2σ	$^{206}Pb/^{238}U$	2σ	$^{207}Pb/^{206}U$	2σ	$^{206}Pb/^{238}U$ (Age) Ma	2σ	$^{207}Pb/^{235}U$ (Age) Ma	2σ	$^{207}Pb/^{206}U$ (Age) Ma	2σ
I1510_1_core	0.66004	0.01316	0.08237	0.00154	0.05812	0.00040	510.2	9.2	514.6	8.0	534.0	14.9
I1510_2_core	13.93067	0.33208	0.52872	0.01218	0.19109	0.00116	2736.0	51.3	2744.9	22.5	2751.5	10.0
I1510_4_core	0.68337	0.01456	0.08505	0.00172	0.05828	0.00039	526.2	10.2	528.8	8.8	540.0	14.7
I1510_5_core	0.67895	0.01416	0.08463	0.00163	0.05818	0.00046	523.7	9.7	526.1	8.5	536.5	17.2
I1510_5_rim	0.67337	0.01325	0.08458	0.00154	0.05774	0.00042	523.4	9.2	522.7	8.0	519.5	16.1
I1510_6_core	0.68059	0.01478	0.08449	0.00171	0.05842	0.00046	522.9	10.2	527.1	8.9	545.5	17.2
I1510_7_rim	0.68067	0.01589	0.08484	0.00188	0.05819	0.00043	524.9	11.2	527.2	9.6	536.5	16.0
I1510_8_core	14.20742	0.29954	0.53718	0.01088	0.19182	0.00112	2771.5	45.5	2763.5	19.9	2757.5	9.6
I1510_8_core2	13.48831	0.27306	0.51318	0.00994	0.19063	0.00113	2670.1	42.3	2714.3	19.0	2747.5	9.7
I1510_9_core	0.68266	0.01421	0.08526	0.00168	0.05807	0.00040	527.4	10.0	528.4	8.6	532.0	15.0
I1510_10_core	0.66020	0.01378	0.08307	0.00164	0.05764	0.00039	514.4	9.8	514.7	8.4	516.0	14.8
I1510_11_rim	0.71942	0.01573	0.08805	0.00182	0.05926	0.00043	544.0	10.8	550.3	9.3	576.5	15.8
I1510_12_core	0.69517	0.01735	0.08603	0.00183	0.05861	0.00077	532.0	10.8	535.9	10.4	552.0	28.7
I1510_13_core	0.77821	0.02613	0.09130	0.00221	0.06182	0.00144	563.2	13.1	584.4	14.9	667.5	49.8
I1510_13_rim	0.68087	0.01449	0.08505	0.00171	0.05806	0.00041	526.2	10.1	527.3	8.7	532.0	15.5

(CL) imaging. The epoxy plug was then washed and sonicated in an HNO_3 solution prior to LA-ICP-MS analysis.

The U-Pb isotopic analyses were conducted on the Nu-Plasma multi-collector plasma source mass spectrometer at the Department of Geological Sciences at the University of Florida. This LA–ICPM–MS is equipped with a custom-designed collector block for simultaneous acquisition of ^{204}Pb (^{204}Hg), ^{206}Pb and ^{207}Pb signals on the ion detectors and ^{235}U plus ^{238}U on the Faraday detectors (Mueller et al., 2008).

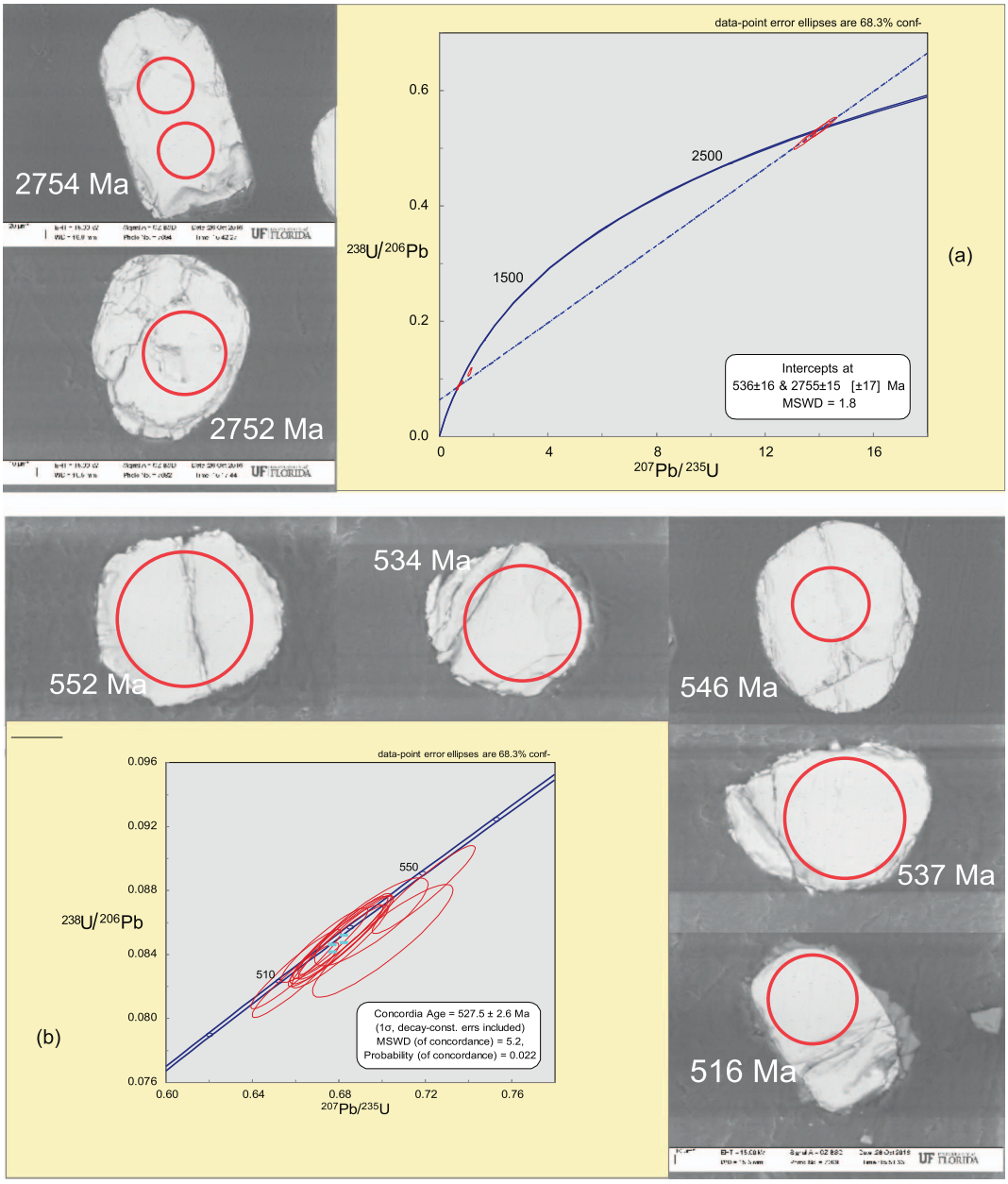


Fig. 3. Electron backscatter imaging and concordia plot of analyzed zircons from site 10. Inherited basement zircon BSE imaging and concordia shown in (a). Inherited ages reflect a phase of Peninsular Gneiss emplacement (2755 ± 15 Ma) and zircons reflecting the crystallization age of dyke T9 (b) also shown, with an age of 527 ± 2.6 Ma. This age represents a relatively late magmatic event as compared to the Proterozoic dykes of the Southern Granulite Terrane. Red circles indicate laser ablation spot size. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Mounted zircon grains were laser ablated using a New-Wave 213-nm ultraviolet laser beam. During U-Pb analyses, the sample was decrepitated in a He stream and then mixed with Ar-gas for induction into the mass spectrometer. Background measurements were performed before each analysis for blank correction and contributions from ^{204}Hg. Each sample was ablated for ~30 s in an effort to minimize pit depth and fractionation following standard practice for zircon analyses at the University of Florida. Data calibration and drift corrections were conducted using the FC-1 Duluth Gabbro zircon standard. Data reduction and correction were conducted using a combination of in-house software and Isoplot (Ludwig, 1999).

3. Results

3.1. Geochronology

3.1.1. Site 10 geochronology

Table 1 presents U-Pb isotopic data on zircons recovered from a mafic dyke sample from Site 10, just southwest of the town of Tiruvannamalai. The sample yielded two concordant zircon ages at 2753 ± 10 Ma (Fig. 3a), consistent with basement ages in the area (Ghosh et al., 2004; Collins et al., 2014). Eleven zircon grains define an age of 527.5 ± 2.6 Ma while a single discordant grain falls at 584 ± 15 Ma (Fig. 3b). Clark et al. (2009) suggest the 525–540 Ma period for collision of the Madurai Block (Azania) with the Salem Block that is consistent with these U-Pb ages (see also Santosh et al., 2017). Site 10 is an abandoned quarry, one of three paleomagnetic sampling

Table 2
Geochronological results for site 35.

Grain name	$^{207}Pb/^{235}U$	2σ	$^{206}Pb/^{238}U$	2σ	$^{207}Pb/^{206}U$	2σ	$^{206}Pb/^{238}U$ (Age) Ma	2σ	$^{207}Pb/^{235}U$ (Age) Ma	2σ	$^{207}Pb/^{206}U$ (Age) Ma	2σ
I1535_1_core	12.16985	0.25440	0.49974	0.00994	0.17662	0.00114	2612.6	42.6	2617.5	19.5	2621.0	10.7
I1535_1_rim	11.90738	0.25137	0.48776	0.00987	0.17706	0.00106	2560.9	42.7	2597.0	19.7	2625.0	10.0
I1535_2_core	11.25309	0.23223	0.47690	0.00935	0.17114	0.00110	2513.6	40.8	2544.2	19.2	2568.5	10.7
I1535_2_rim	11.15138	0.23583	0.47794	0.00948	0.16922	0.00124	2518.2	41.3	2535.7	19.6	2549.5	12.3
I1535_3_core	11.47819	0.23160	0.48937	0.00945	0.17011	0.00100	2567.8	40.8	2562.7	18.8	2558.5	9.8
I1535_3_core2	11.45672	0.25291	0.48889	0.01041	0.16996	0.00099	2565.8	45.0	2560.9	20.5	2557.0	9.7
I1535_5_core	11.68306	0.29429	0.49492	0.01208	0.01712	0.00108	2591.8	52.0	2579.2	23.4	2569.0	10.5
I1535_6_core	9.34457	0.13934	0.44735	0.00614	0.15150	0.00089	2383.4	27.3	2372.3	13.6	2362.5	10.0
I1535_7_core	9.46768	0.17983	0.44419	0.00677	0.15459	0.00175	2369.3	30.2	2384.3	17.4	2397.0	19.3
I1535_8_core	9.16364	0.13302	0.44125	0.00589	0.15062	0.00086	2356.2	26.3	2354.4	13.2	2352.5	9.7

locations on a NW-SE trending dyke in the Tiruvannamalai area. The characteristic remanence of this dyke (D = 115°, $I = -70°$; k = 98, $\alpha_{95} = 3°$) was consistently overprinted by a northerly and shallow component (D = 16°, $I = -17°$; k = 10, $\alpha_{95} = 16°$). This dyke is cut by a number of subparallel NE-SW trending dykes, including dyke T9 of Radhakrishna and Joseph (1996). One of these NE-SW cross-cutting dykes was observed in the abandoned quarry at site 10 but was unreachable due to high water levels. Given the discrepancy between the paleomagnetic data at site 10 and the geochronologic data, it is likely that the 527 ± 2.6 Ma age dates the crystallization age of the cross-cutting NE-SW dykes. This age therefore delineates limited Cambrian magmatism in the Tiruvannamalai area (see discussion for details).

3.1.2. Site 35 geochronology

Table 2 lists U-Pb isotopic data for zircon separates from the Site 35 dyke near Panchalli. Samples from this site yielded six concordant analyses from 3 inherited zircon grains (2755 ± 15 Ma) that correspond with basement rocks in the region (Ghosh et al., 2004; Collins et al., 2014). In addition, they yield 3 concordant analyses from 3 zircon grains that define an age of 2363 ± 6.6 Ma that we attribute to the crystallization of the dyke (Fig. 4). Paleomagnetic data from this NW-SE trending dyke yield a direction of D = 61°, $I = -84°$; k = 169, $\alpha_{95} = 4°$.

3.2. Paleomagnetism

Paleomagnetic results from the dykes of the Southern Granulite Terrane fall into two main groups (accounting for 50% of sites in this study) consistent with previously reported directions (Belica et al., 2014; Dash et al., 2013; Halls et al., 2007; Kumar et al., 2012a,b; Meert et al., 2011; Pisarevsky et al., 2013; Radhakrishna and Joseph, 1996; Venkatesh et al., 1987). In the following discussion, we refer to positive inclinations as 'normal' polarity and negative inclinations as 'reverse' polarity.

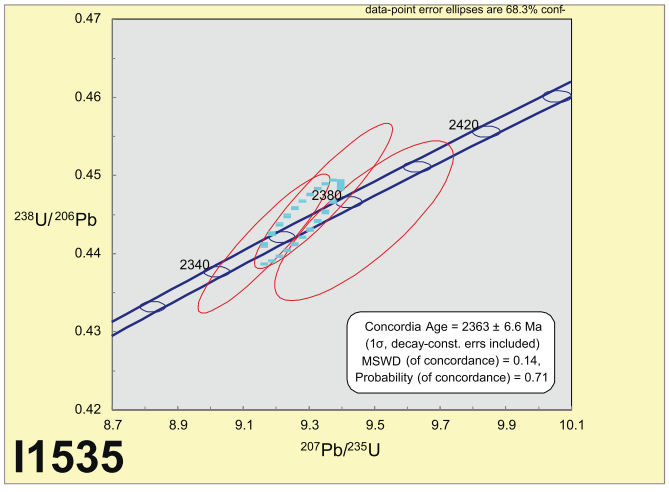


Fig. 4. Concordia plot from zircons recording the crystallization age of the dyke at site 35, with an age of 2363 ± 6.6 Ma. This age is a precise match with the giant Dharwar dyke swarm dated by Halls et al. (2007) at 2367 ± 1 Ma.

3.2.1. Low to moderate temperature/coercivity components

Low/intermediate temperature and coercivity components were removed by 500 °C or 15–60 mT. The majority of the low-intermediate temperature/coercivity directions yielded shallow northerly directions with D = 353°, $I = -5°$; k = 8, $\alpha_{95} = 16°$ (12 sites; Fig. 7e). This overprint is similar to the "B" component of Halls et al. (2007). However, some well-grouped secondary components were distinctly different from this more common direction. Sites 22 and 31 (each on a separate dyke) share the same overprint direction of D = 102°, I = +1°; $\alpha_{95} = 50°$, k = 27. This overprint is close to a high temperature component seen in the basal Marwar Supergroup (of late Ediacaran age) with D = 89°, $I = -1.0°$; $\alpha_{95} = 9°$ (Davis et al., 2014). It is also comparable to the paleomagnetic direction recovered at site T9 (Radhakrishna and Joseph, 1996).

3.2.2. Steeply inclined directions

Following the removal of lower temperature and coercivity components, the most prevalent paleomagnetic direction characteristic of 17 dykes (20 sites) across our study area was a relatively steep dual-polarity remanence isolated at high temperatures or high applied fields (Table 3; Fig. 7). Some sites showed univectorial decay (Fig. 7c). The remanence is consistent with prior results on dykes dated to 2370 Ma in the SGT/Dharwar craton west of our sampling area (Halls et al., 2007) and north of our sampling area in the Dharwar craton "proper" (Belica et al., 2014) and 2318 Ma in the Southern Granulite Terrane (Radhakrishna et al., 2013; Dash et al., 2013). The 17 reversed sites (14 dykes) had a mean D = 28°, $I = -84°$ (k = 31; $\alpha_{95} = 7°$) while the 3 normal sites (3 dykes) are less well grouped at D = 277°, I = +72° (k = 21; $\alpha_{95} = 28°$; Fig. 7d). A reversal test (McFadden and McElhinney, 1990) between the two directions (with normal polarity dykes P69, GR, and GT from Belica et al. (2014) added to the normal polarity data) passed the test with a classification of R_c ($\gamma_c = 15.3°$ and a $\gamma_{obs} = 4.3°$).

A baked contact test was conducted at Sites 30a,b and 31 (Fig. 2; Table 6; Fig. 8i). These samples were collected about 1 km apart on the same ~E-W trending dyke (Fig. 2). The main dyke at site 30a is intruded by a smaller, fine-grained, subparallel mafic dyke that is not present at site 31 (Fig. 8a). Directional data from site 31 show two components of magnetization. The first, a low-coercivity component (< 10 mT), is somewhat scattered but forms a reasonable grouping to the SE-shallow (D = 110°, I = +15° (k = 14, $\alpha_{95} = 11°$) and a higher coercivity direction with a mean D = 342°, $I = -68°$ (k = 146, $\alpha_{95} = 4°$; Fig. 8b).

The magnetic signature at site 30a,b is more complex due to the presence of the smaller dyke. Samples from the smaller dyke at site 30b show a consistent WNW shallow to intermediate upwardly directed

Table 3
Paleomagnetic results from 2.37 Ga dykes in the Southern Granulite Terrane.

Site	Slat (°)	Slong (°)	Trend	B/N	D (°)	I (°)	α95 (°)	k	Plat (°N)	Plong (°E)	A95	Refs.
2	13.6928	78.9235	20	9	266	−65	2	630	13	124		This study
4	13.6737	79.1513	70	7	94	58	3	160	6	130		This study
7	13.0867	78.9748	70	9	190	−83	3	209	25	81		This study
10 + 11 + 12	12.1773	78.9301	120	3/22	115	−70	3	96	24	43		This study
15	12.1585	79.1736	130	7	278	−78	4	145	8	103		This study
22	12.0767	77.9324	90	7	89	−81	13	20	11	60		This study
28	12.0761	77.8397	165	5	50	−83	5	159	3	67		This study
29	12.0620	77.8343	130	6	13	−75	7	87	−15	72		This study
30i + 31	12.0505	77.8224	140	2/17	350	−68	4	103	−26	85		This study
32	12.0637	77.8072	135	7	114	−80	1	1573	19	60		This study
33	12.4546	78.3539	80	7	51	−76	8	90	−5	57		This study
34*	12.4523	78.3972	130	8	16	−74	5	125	−28	64		This study
35	12.4764	77.9656	150	7	61	−84	4	169	7	68		This study
37	12.4716	77.8552	135	7	179	81	4	166	−5	79		This study
38	12.4692	77.8570	130	7	19	−70	5	121	−21	70		This study
39	12.4599	77.8429	100	7	42	−76	3	176	−8	60		This study
40	12.4839	77.8152	90	7	285	68	3	228	19	39		This study
Mean				17/152	49	−83	7	27	2	74	14	This study
Tiruv.			NW-SE	10	125	−74	8	22	28	52	13	1
Tiruv.			NW-SE	7	107	−78	9	42	19	55	–	2
C				4	133	−75	6	134	31	54	11	3
2.37				18	65	−81.7	8.3	19	6.6	63.1	8.3	4
“A”			*E*-W	7	94.9	−80.9	7.3	69	13.6	59.6	13.9	5

Each table entry (except for means) represents one dyke, if multiple sites are in the same row, they came from the same dyke. Slat = site latitude, Slong = site longitude, Trend = trend of dyke, B = number of sites (or number of dykes for mean directions), N = number of specimens, D = declination, I = inclination, a95 = cone of 95% confidence about the mean direction, k = kappa precision parameter (Fisher, 1953), Plat = pole latitude, Plong = pole longitude, A95 = radius of the 95% confidence circle about the mean pole, References: 1 = Dash et al. (2013); 2 = Radhakrishna et al. (1996); 3 = Venkatesh et al. (1987); 4 = Belica et al. (2014); 5 = Halls et al. (2007).

magnetization with a mean D = 284°, $I = -25°$ (k = 375, $\alpha_{95} = 5°$; Fig. 8d) that is nearly antipodal to the overprint at Site 31. Samples from the interior portion of the larger E-W trending dyke at site 30a reveal a multicomponent magnetization. A shallow to intermediate upwardly directed magnetization is unblocked at lower applied fields/temperatures with a mean at D = 333°, $I = -33°$ (k = 33, $\alpha_{95} = 10°$; Fig. 8c). A direction comparable to site 31 is identified at higher temperatures and coercivities with a mean D = 360°, $I = -67°$ (k = 94, $\alpha_{95} = 6°$; Fig. 8c).

A gneissic sample adjacent to the smaller dyke at site 30b shows a direction indistinguishable from the smaller dyke (Fig. 8e). Outside of the baked zone from the smaller dyke (≫0.5 dyke width, 30b), but still within the baked zone of the main dyke at site 30a (≪0.5 dyke width); samples show steeply inclined upwardly directed magnetizations identical to those at site 30a and 31 (Fig. 8f,g). Gneissic samples distant from the contacts of both dykes show intermediate NE-downwardly directed magnetizations (Fig. 8h). These directions are identical to the stable mean direction obtained at basement reference site 46 (Table 6).

These data suggest that dyke 30b (the smaller dyke) emplaced along the margin of dyke 30a and baked the gneissic samples along the margin of dyke 3b and overprinted samples at site 30a. Furthermore, the larger dyke at site 30a baked gneissic samples located ≫0.5 dyke width from dyke 30b. Gneissic samples outside the expected baked zone of both dykes show a unique and stable direction. We consider this as a positive baked contact test for dykes 30a and 31. This unique geometry also provides an inverse baked contact test for site 30b on dyke 30a. Finally, a positive baked contact test for the smaller dyke 30b is evident by the complete overprinting of the adjacent gneissic samples and the partial overprinting of the larger dyke.

Data from Curie temperature experiments show reversible heating/cooling curves indicating relatively unaltered magnetite as the primary magnetic carrier in all dykes of this group (Fig. 5a). All dykes showed either PSD or a SD/PSD mixed behavior on a Day Plot (Fig. 6).

The mean paleomagnetic pole for the steep directions across the Southern Granulite Terrane (17 dykes) falls at 2°N, 73°E (A95 = 14°). Our results in the Southern Granulite Terrane come from more geographically dispersed dykes than those reported in Dash et al. (2013). That would indicate that dykes of the 2.37 Ga swarm cover a wide area within the SGT. We note that our findings are consistent with the directional bias towards reverse polarity seen in previous studies wherein 83% of our results show ‘reverse’ polarity (Halls et al., 2007; Dash et al., 2013; Belica et al., 2014; this study).

3.2.3. *Shallowly inclined directions*

We also identify a north-south and shallowly inclined dual polarity magnetization (Table 4; Fig. 9c). This magnetization was interpreted by Halls et al. (2007) as an Ediacaran-Cambrian overprint. While this direction was most commonly observed as a low-temperature overprint in dykes from the Dharwar craton, it is the only well-resolved component from 12 dykes in our study. Multiple dykes showed only this direction (Fig. 9a) while others (Fig. 9b) showed less-well resolved overprints. Our results yield a mean ‘normal’ polarity with D = 353°, I = +5° (k = 16, $\alpha_{95} = 13°$; $N = 10$) and a mean ‘reverse’ polarity group with D = 199°, $I = -16°$ (k = 12, $\alpha_{95} = 38°$; $N = 3$).

A baked contact test was attempted at site 18 near Attur, Tamil Nadu (Fig. 10). Samples from the interior and margin of the dyke yielded a mean D = 11°, I = +15° (k = 82; $\alpha_{95} = 4°$; Fig. 10b). This shallow, northerly direction suggests a complete remagnetization of this dyke. Three samples were taken from within 45 cm of the dyke delineating a scattered mean D = 352°, $I = -21°$ ($\alpha_{95} = 40°$; Fig. 10c) while samples taken 5 m from the dyke-host contact showed an independent group of directions at D = 117°, $I = -42°$ ($\alpha_{95} = 17°$; Fig. 10d). Given the size of this dyke (~30 m), the group at 5 m should still be well within the partial bake zone; however, the paleomagnetic directions of the host rock do not correspond with that of the dyke (Fig. 10a,e). This means that the baked contact test fails (as is expected given the remagnetization of the dyke), but the host rock is not remagnetized in the same manner. The transitional nature of the basement directions with distance, roughly falling along a great circle path towards steeper directions (Fig. 10e), may reflect the cryptic survival of the original “baked” magnetization in the host rock, now entirely overlapped by the overprinted magnetization. Since site 18 is our southernmost dyke, and south of the Salem-Attur Shear Zone, a complete overprinting of its directions is not surprising.

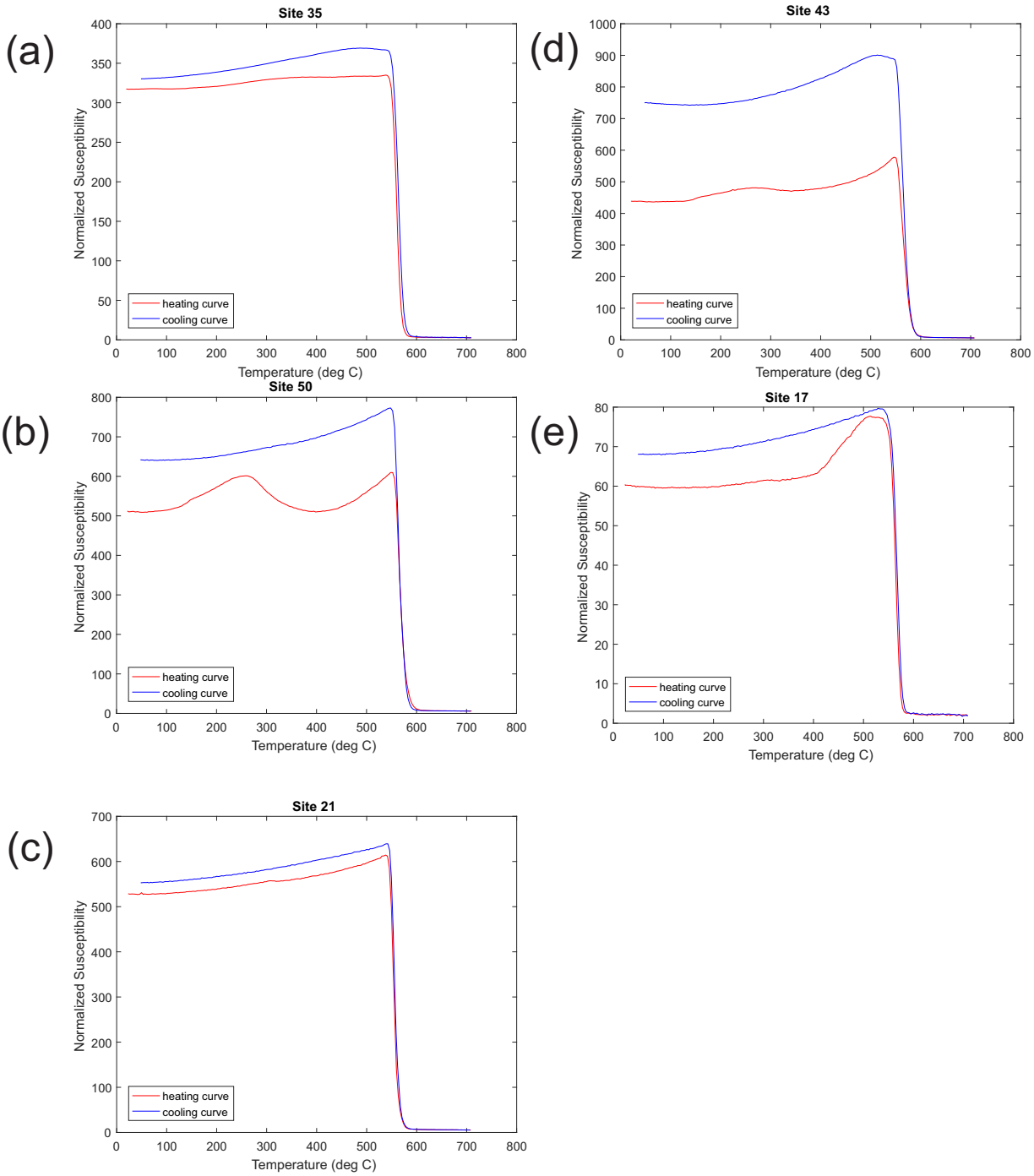


Fig. 5. Susceptibility vs. temperature curves for selected sites from the recovered paleomagnetic directional groupings: (a) from the 2.37 Ga direction exhibiting a reversible heating cooling curve, this dyke was also dated (b) possibly from the 2.21 Ga direction, prominent pyrrhotite bump at 320 °C (c) from an undated direction, reversible heating cooling curve (d) dyke overprinted in the Neoproterozoic, changes upon heating indicating alteration (e) dyke associated with the 1.88 Ga direction, abrupt peak leading up to magnetite Curie temperature.

Curie temperature curves for these dykes generally showed lower susceptibility than primary dykes, and the heating/cooling curve disparity indicates alteration, although magnetite is still the unambiguous magnetic signal carrier (Fig. 5d).

The overall mean pole position for the thirteen shallow inclination directions from these dykes falls at 82° N, 267° E (A95 = 11°). Halls et al. (2007) reported a mean pole position of 79° N, 257° E (A95 = 6°) that was argued to represent an Ediacaran overprint. This paleomagnetic direction was also used to calculate the paleomagnetic pole of the Dharmapuri "swarm" by Radhakrishna and Joseph (1996), with a pole falling at 80° N, 259° E although they assigned an age of 1800–1750 Ma to this swarm based on their prior work (Radhakrishna et al., 1999).

3.2.4. *Dual polarity intermediate directions*

A consistent dual-polarity paleomagnetic signature with a northwest/southeast declination and intermediate positive/negative inclinations (Table 5C; Fig. 11c) was observed in 8 dykes (8 sites) throughout the study area (Table 5B). At site 1, this direction was recovered from the dyke as well as baked gneiss host, although another baked contact test at site 21 was inconclusive. Stepwise

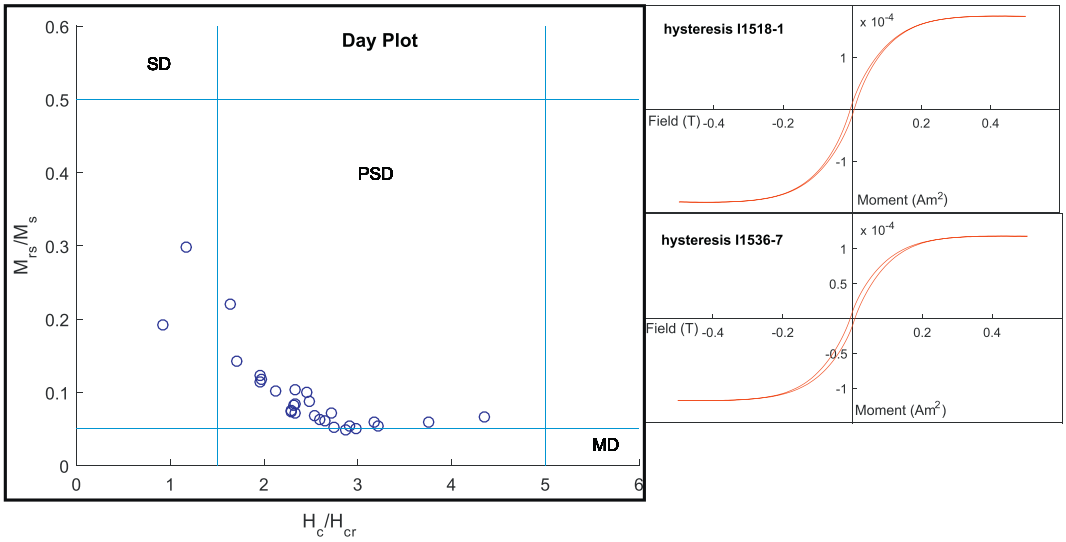


Fig. 6. Day Plot (Day et al., 1977) showing magnetic domain characteristics from a representative suite of dyke samples throughout the Southern Granulite Terrane. They primarily exhibit pseudo-single domain magnetite behavior. Hysteresis loops from selected samples are also shown, also consistent with magnetite.

demagnetization showed the remanence as bivectorial with mostly present-day field overprints at lower temperatures/applied fields (11a,b). The mean direction from the 3 'reverse' polarity sites is D = 140°, $I = -54°$ (k = 34; $\alpha_{95} = 22°$). The normal polarity mean direction is D = 332°, I = +53° (k = 37; $\alpha_{95} = 13°$). This direction (normal polarity) was also seen in Lakhna dyke D4 in the Bastar Craton by Pisarevsky et al. (2013), but excluded from their 1.47 Ga mean. Curie temperature experiments show magnetite to be the primary magnetic carrier, with reversible heating/cooling curves showing little alteration (Fig. 5c). Dykes carrying this remanence trend N-S or ~NW-SE in the SGT and E-W in the Dharwar Craton.

Five dykes of this grouping were of normal polarity and three dykes were of reverse polarity. The mean VGP for this direction is at 53° N and 32° E (A95 = 11°). A reversals test (McFadden and McElhinney, 1990) between the two directions is classified as R_c ($\gamma_c = 18.1°$, $\gamma_{obs} = 11.7°$). This test included one dyke of reverse polarity direction that was excluded from the Tiruvannamalai mean by Radhakrishna and Joseph (1996) to bolster the reverse polarity dataset. This direction is less prevalent in the Southern Granulite Terrane as compared to the 2.37 Ga direction and the shallow overprint direction.

3.2.5. Other directions

Two dykes from this study carried antipodal paleomagnetic directions (Table 5A; Fig. 12c), consistent with the large 2.21 Ga swarm dated by Kumar et al. (2012a). These dykes trended NE-SW or E-W, which is different than the typical trend (N-S) for this swarm in the northern Dharwar craton. Both dykes showed relatively univectorial decay during stepwise thermal demagnetization (12a,b). The mean paleomagnetic directions for each site were roughly antipodal (Fig. 12c) with a mean 'normal' polarity D = 57°, I = +41° (k = 45; $\alpha_{95} = 6°$) and a mean 'reverse' polarity of D = 238°, $I = -64°$ (k = 264, $\alpha_{95} = 3°$). Curie temperature runs on these dykes confirmed magnetite as the primary carrier, consistent with observations during demagnetization of the specimens (Fig. 5b). The heating curve near 320 °C shows the presence of minor pyrrhotite in site 50 (Fig. 5b). The mean VGP for these dykes was at 34° S and 316° E which is similar to both the Kumar et al. (2012a, b) pole at 32° S and 302° E (A95 = 9°) and the Belica et al. (2014) pole at 30° S and 297° E (A95 = 22°). Dykes of this age are uncommon in the Southern Granulite Terrane. Our results seem to indicate that the 2.21 Ga swarm is present throughout the entire Dharwar Craton, but not as densely emplaced as the 2.37 Ga

Table 4
Paleomagnetic results from Southern Granulite Terrane dykes overprinted in the Ediacaran.

Site	Slat (°)	Slong (°)	Trend	B/N	D (°)	I (°)	α95 (°)	k	Plat (°N)	Plong (°E)	A95	Refs.
3	13.6979	78.9215	80	7	333	9	8	34	62	333		This study
6	13.0883	78.9849	0	7	342	−11	7	51	65	305		This study
8	13.0557	79.0384	90	7	340	0	4	138	62	320		This study
18i + m	11.6576	78.4911	30	12	11	15	4	83	81	174		This study
19	12.1513	78.2554	130	9	190	−21	5	83	87	145		This study
20	12.1306	78.3189	140	8	218	−4	7	46	76	276		This study
24	12.1461	78.2666	60	9	183	−26	2	236	87	275		This study
25	12.0708	78.3834	140	9	356	−4	3	106	67	345		This study
27	12.0778	77.8369	165	3	359	18	10	89	78	196		This study
42	12.4647	77.7673	160	8	336	21	3	170	68	226		This study
43	12.1873	77.7391	60	12	11	13	6	46	79	190		This study
44	12.1229	77.8405	40	7	12	−13	4	102	51	180		This study
Mean				12/99	359	9	13	13	82	267	11	This study
Grand Mean				24 dykes					81	259	6	This study
Ag-Anaik.				2	14	27	26	94	76	153		1
Dharm.				5	359.8	9.2	10.8	51	79.7	281.1		1
'B'												2

Slat = site latitude, Slong = site longitude, Trend = trend of dyke, B = number of sites (or number of dykes for mean directions), N = number of specimens, D = declination, I = inclination, a95 = cone of 95% confidence about the mean direction, k = kappa precision parameter (Fisher, 1953), Plat = pole latitude, Plong = pole longitude, A95 = radius of the 95% confidence circle about the mean pole, References: 1 = Radhakrishna et al. (1996); 2 = Halls et al. (2007).

Table 5
Grouped paleomagnetic results of uncertain age from the Southern Granulite Terrane (likely 2.21 Ga to 1.88 Ga).

Site	Slat (°)	Slong (°)	Trend	B/N	D (°)	I (°)	α95 (°)	k	Plat (°N)	Plong (°E)	A95	Refs.
5c: 2.21 Ga dykes?												
5	13.0863	79.1369	70	9	57	41	6	45	35	150		This study
50	12.6458	77.8420	100	7	238	−63	3	264	32	124		This study
2.21			N-S	6/39	236	−67	20	12	32	117	12	1
N-S dykes			N-S	3	239	−64	6	43	32	122	9	2
5b: 1.88 Ga dykes?												
1b	13.65262	78.96585	80	2	283	−7	13	54	−12	163		This study
17	12.25992	78.9639	60	5	306	−21	12	44	−32	148		This study
30 m	12.0495	77.8276	30	4	284	−25	5	377	−11	153		This study
Mean				3/10	291	−17	23	30	18	334	21	This study
1.88				29	129	9	7	19	36	331	7	1
Bastar				7	126	15	12	27	31	330	12	3
5c: NW-SE intermediate directions												
1a	13.65262	78.96585	80	5	320	41	13	16	53	11		This study
9	12.06762	79.00718	70	8	142	−68	2	260	40	49		This study
14	12.28115	78.8901	105	9	131	−49	5	87	41	18		This study
21	11.91903	78.38728	130	7	341	67	6	43	49	56		This study
23	12.11713	77.83998	60	6	147	−44	4	95	57	17		This study
36	12.40667	77.96142	170	7	322	55	3	162	48	29		This study
45	12.60592	78.11583	20	7	326	56	5	113	51	32		This study
47	12.61783	77.7131	135	5	354	44	13	38	76	55		This study
Mean				8/54	327.7	53.9	9.3	36.5	52.9	32.2	10.6	This study

Each table entry represents one dyke, if multiple sites are in the same row, they came from the same dyke. Slat = site latitude, Slong = site longitude, Trend = trend of dyke, B = number of sites (or dykes for mean directions), N = number of specimens, D = declination, I = inclination, a95 = cone of 95% confidence about the mean direction, k = kappa precision parameter (Fisher, 1953), Plat = pole latitude, Plong = pole longitude, A95 = radius of the 95% confidence circle about the mean pole, References: 1 = Belica et al. (2014); 2 = Kumar et al. (2012a, b); 3 = Meert et al. (2011).

Table 6
Directional paleomagnetic results from a baked contact test at sites 30a,b and 31.

Site	Slat (°)	Slong (°)	Trend	B/N	D (°)	I (°)	α95 (°)	k
30a	12.0495	77.82763	110	10	360	−67	6	94
31	12.05145	77.81717	110	7	342	−68	4	146
Baked gneiss	12.0495	77.82763	110	2	348	−78	42	10
30b	12.0495	77.82763	110	4	284	−25	5	375
Baked gneiss	12.0495	77.82763	110	1	283	−22	N/A	N/A
Unbaked gneiss	12.0495	77.82763	110	4	44.3	39	15	40
46	12.64368	77.74318	N/A	10	24.1	35	9	30

Slat = site latitude, Slong = site longitude, Trend = trend of dyke, B = number of sites, N = number of specimens, D = declination, I = inclination, a95 = cone of 95% confidence, k = kappa precision parameter (Fisher, 1953).

swarm.

Three sites in this study carry paleomagnetic signatures consistent with the reverse polarity of the widespread 1.88 Ga large igneous province (Table 5b; Fig. 12e). Two of these dykes are from sites 1 and 30b which exhibit multicomponent magnetizations (Figs. 7, 8, 13). At site 31, a scattered overprint (D = 110°, I = +15° (k = 14, α_{95} = 11°) falls roughly antipodal to this paleomagnetic direction. Such directions at the margin of larger, older dykes imply that emplacement of the 1.88 Ga swarm in the Southern Granulite Terrane was along the same zones of weakness as the older swarms. Stepwise demagnetization revealed bivectorial demagnetization paths, with lower temperature components being removed by 500 °C (Fig. 12d). Curie temperature analysis confirms magnetite to be the primary signal carrier (Fig. 5e). The mean direction of these sites is D = 291°, $I = -17°$ (k = 30;

Table 7
Comparative Gondwanan poles from the Ediacaran-Cambrian.

Unit name	Abbreviation	Region	Age (Ma)	Plat (°N)	Plong (°E)	A95	Weight	Ref.
Brachina Formation	BFm	Australia-S	603	−33	328	16	B	McWilliams and McElhinny (1980)
Baganpalli quartzite	BQ	India	589	−74	54	4	B	Goutham et al. (2006)
“Shallow” overprint	SO	South India	Ediacaran	82	267	11	B	This study
Sierra de las Animas complex C	SdlA	Rio de la Plata	578	−12	259	14	B	Rapalini et al. (2015)
Nola metadolerite	NMD	Congo	571	−62	305	8	B	Moloto-A-Kenguemba et al. (2008)
Adrar-n-Takoucht	A-n-T	West Africa	570	−57	296	16	B	Robert et al. (2017)
Lower Arumbera Formation	LA	Australia-N	561	−44	342	10	A	Kirschvink (1978)
Tadoughast	Tad	West Africa	561	22	31	16	B	Robert et al. (2017)
Sinyai metadolerite	SMD	Congo	547	−29	319	4	A	Meert and Van der Voo (1996)
Marwar Supergroup	Marwar	NW India	late Ediacaran	−1	344	7	B	Davis et al. (2009)
Dichol-Boho	D-B	West Africa	536	23	27	15	B	Robert et al. (2017)
Dyke T9*	T9*	South India	527?	7	347	5	C	This study
East Coast Dykes	ECD	South India	527?	2	8	6	B	Dash et al., (2013)
SGT overprints*	SGTO*	South India	527?	7	352	12	C	This study
Stratoid granite	SG	Madagascar	520	−7	353	14	A	Meert et al. (2003)
Carion granite	CG	Madagascar	508	−7	1	15	A	Meert et al. (2001)

Abbreviations are those used in Fig. 17. Ages are nominal averages of uncertainty. Poles are not rotated in this compilation, but were rotated into an African reference frame for comparative purposes (see Fig. 17) using the following Euler rotations. MAD-AFR: -3.41, −81.7, 19.73; IND-AFR: 27.9, 43.64, −64.4; AUS-AFR: 24.5, 112.3, −56.3; SA-AFR: 45.5, −32.2, 58.2. Weight is a measure of confidence in the result, with ‘A’ being highest confidence.

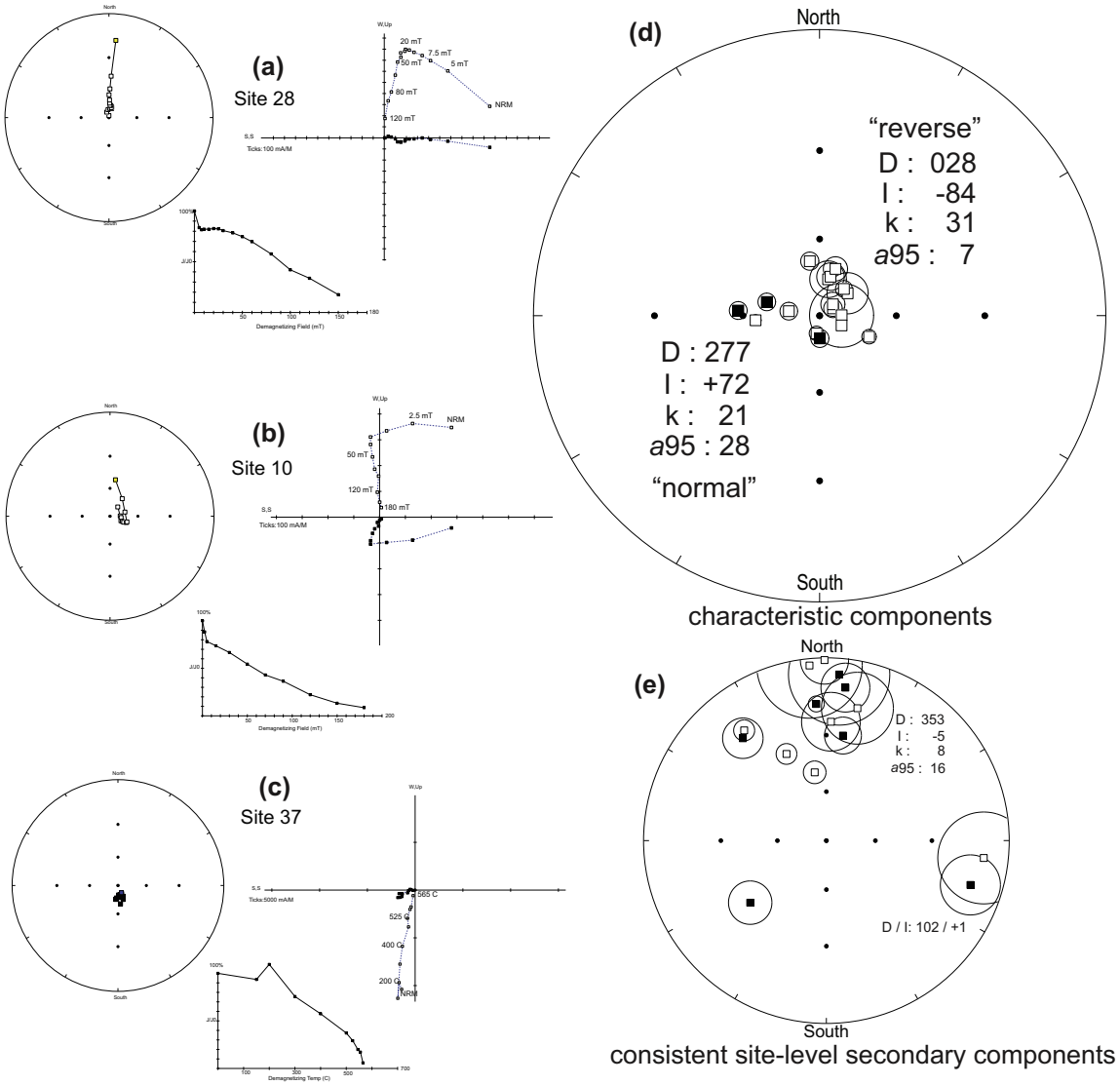


Fig. 7. Demagnetization data for reverse (a,b) and normal polarity (c) sites from 2370 Ma dykes in the Southern Dharwar craton and northern Southern Granulite Terrane. Note that (a,b) show overprinting consistent with the "B-component" discussed in Section 3.2.3 and shown in Fig. 9. Stereoplots show characteristic site mean directions (d) and (e) consistently grouped low-temperature/coercivity components.

α_{95} = 23°).

After inverting the 'reverse' polarity sites, the mean VGP falls at 18° N and 334° E (A95 = 21°). This pole would indicate a higher latitude position for India than calculated from the Meert et al. (2013) pole at 31° N and 330° E (A95 = 12°) or the Belica et al. (2014) pole at 37° N and 334° E (A95 = 6°). However, given the limited nature of 1.88 dyke outcroppings in this area, we believe this is due to inadequate averaging of paleosecular variation in our VGP. Indeed, our 3 dyke directions and the VGPs calculated from them are consistent with individual VGPs from both Meert et al. (2013) and Belica et al. (2014).

4. Discussion

4.1. Paleomagnetism at scale

It is apparent from the preponderance of several dyke systems in peninsular India that assigning ages to dykes based on trends, while demanded from a logistical standpoint, may not always be accurate. It is nearly impossible to visit every single dyke in a given area – and not all are suitable for paleomagnetism and geochronology. A zone of weakness (structural grain/fractures, etc.) exploited by an early generation of dykes can remain as an efficient propagation locale for a subsequent generation of dykes. Indeed, our work in the southern Dharwar/Southern Granulite Terrane presents multiple examples of disparate data in close proximity. The site 30 double-baked contact tests discussed earlier were an excellent example, but other sites showed similarly complex overprints and primary directions.

Site 1, near Pileru, displays at least two generations of dykes. The site was on the second level of a small roadside quarry. Samples were taken from fine-grained dyke at the contact with granitic gneiss host near the contact and from small (ca. 2 cm wide) dykes hosted by the granite away from the contact. These small dykes in the granite from the mm to cm scale are quite abundant at this site. Samples were taken from west to east down the outcrop, with samples 1–9 in the west and 10–12–50–70 m to the east. The more westerly samples (10 specimens) defined a northwesterly intermediate direction D = 320°, I = +41° (α_{95} = 13°) while the easterly samples (4 specimens) defined a direction D = 283°, *I* = −7° (α_{95} = 13°; Fig. 13a). Satellite imagery of the area shows both *E*-W and NNE-SSW dykes cross-cutting in the vicinity of site 1.

Site 27 samples were taken from small "dykelets" that branch far from the main dyke at site 28. While the main dyke at site 28 has a

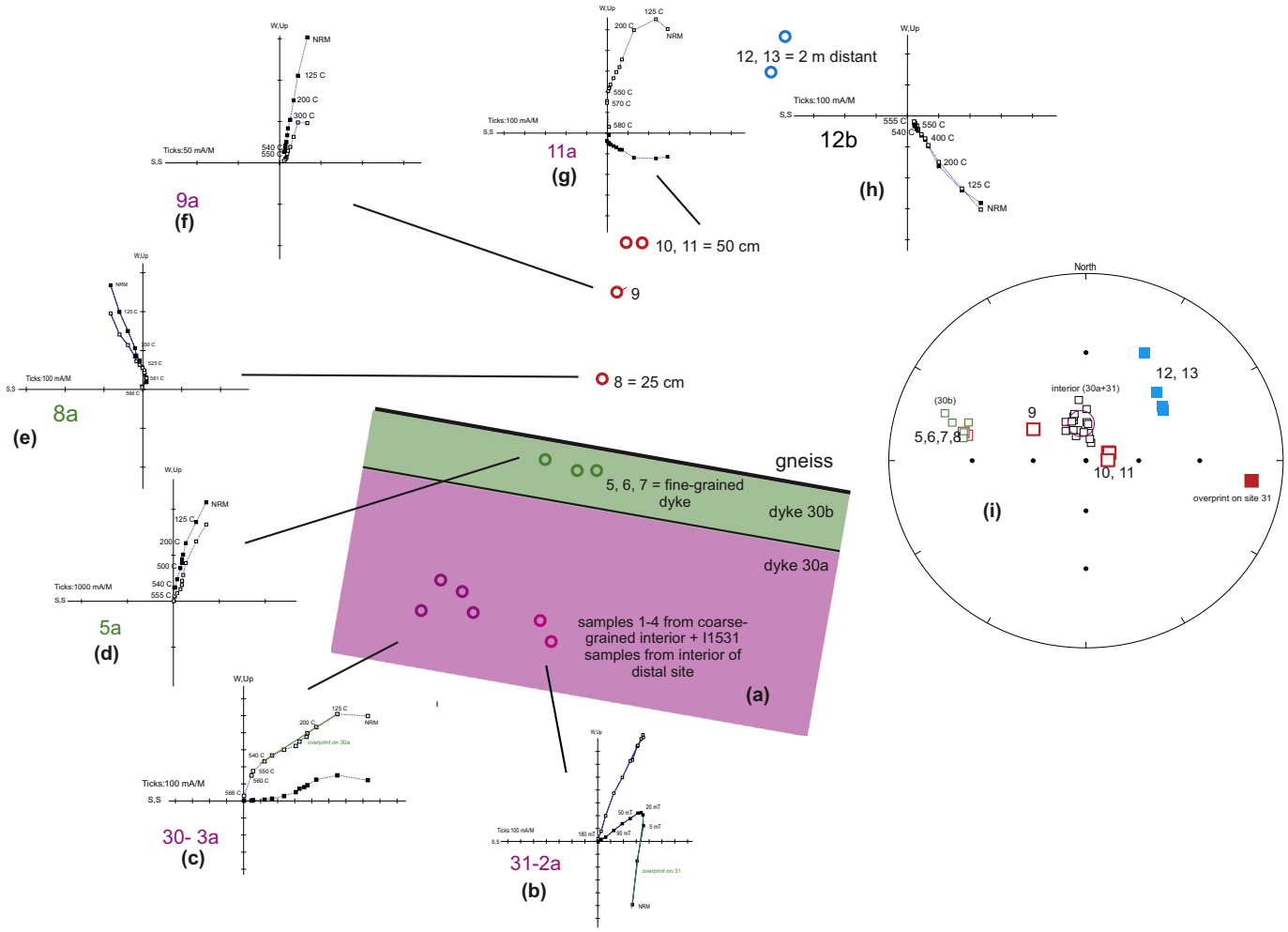


Fig. 8. Map view schematic of baked contact test at site 30 (a), along with demagnetization data. Demagnetization data shown is from a distal site on the same dyke (site 31, note overprint; b), coarse-grained large dyke (site 30a; c), fine-grained marginal dyke (site 30b; c), and the baked zone to unbaked zone transition (e-h). Stereoplot with directional data also shown (i). These sites provide evidence for two primary remanences: the 2.37 Ga direction and the 1.88 Ga direction. This rather unusual situation underscores the complexities of terranes with multiple phases of dyke emplacement.

steep negative inclination (D = 50°, $I = -84°$; $\alpha_{95} = 5°$), site 27 dykelets show a shallow northerly inclination (D = 359°, I = +18°; $\alpha_{95} = 16°$). These two directions are comparable, respectively, to the canonical 2.37 Ga direction of the giant "Bangalore" swarm (Halls et al., 2007; Belica et al., 2014) and the B directions of Halls et al. (2007), the latter inferred to be an Ediacaran overprint. It seems likely that the interior of the dyke managed to retain the primary remanence while the dykelets were completely overprinted. This interpretation is supported by the fact that the low coercivity (to 30 mT) remanence from site 28 is identical to the overprinted site 27 direction (Fig. 13b).

4.2. The South Indian "bar code"

Refinement in age control through more precise age data has changed the paradigm for Paleoproterozoic dyke emplacement in the South Indian Block from relatively few episodes to several (Fig. 14). The giant 2370 Ma "Bangalore" dyke swarm has been unequivocally dated throughout the Dharwar Craton by multiple workers (Halls et al., 2007; French et al., 2008; Fig. 4, this study). Consistent paleomagnetic data from this dyke swarm (Halls et al., 2007; Belica et al., 2014, Dash et al., 2013; Radhakrishna and Joseph, 1996, Radhakrishna et al., 2013; Venkatesh et al., 1987, this study) are similarly widespread, from the Dharwar Craton into the Salem Block. These poles are more scattered in the Salem Block than noted elsewhere, perhaps reflecting greater tectonic reworking of this block as compared to the more stable Dharwar Craton. A comparison of poles presented in multiple paleomagnetic studies throughout Southern India (Fig. 15) shows relatively good agreement between paleomagnetic poles from this dyke swarm. As noted before, most of these data show a reverse polarity, but normal polarity dykes belonging to this swarm have also been identified (Belica et al., 2014, this study). Dash et al. (2013) also noted a normal polarity dyke, as mentioned earlier, yet chose to discard it. Since there are several more 'normal' polarity results that pass a reversal test (Belica et al., 2014, this study) we see no reason to either exclude them from the 2370 Ma paleomagnetic compilation for the Dharwar craton or to regard this swarm as containing only a single polarity. We note that there is a slight discrepancy among the poles presented (Fig. 15). Data primarily from the Southern Dharwar show good agreement, while Southern Granulite Terrane results may indicate some rotation relative to more stable areas of the Dharwar Craton. The Salem Block is dissected by several crustal-scale shear zones (Collins et al., 2014; Fig. 2) which may facilitate small-scale rotations of paleomagnetic data from areas of the SGT. The Dash et al. (2013) results from the 2370 Ma swarm are all from the Tiruvannamalai area, which lies east of one of these shear zones, which may provide a mechanism for the rotations. The notion of rotations in the SGT is admittedly speculative, but worth considering in further evaluation of paleomagnetic data from this area.

Demirer (2012) reported U-Pb baddeleyite ages of 2255 ± 5 Ma

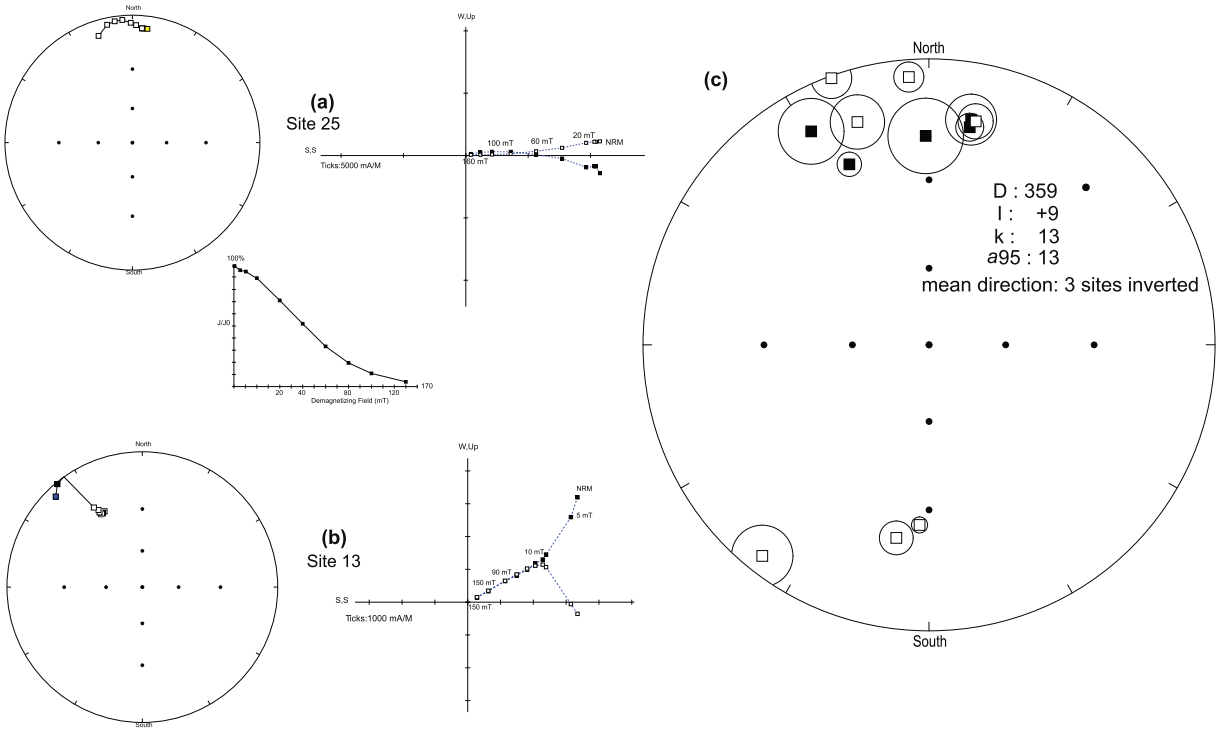


Fig. 9. Demagnetization data for 'B-component' dykes with minor overprints (a,b) in the southern Dharwar and Southern Granulite Terrane interpreted to be products of Ediacaran remagnetization. Shallow paleomagnetic directions indicative of this grouping (c), also corresponding with consistent overprints on older paleomagnetic directions.

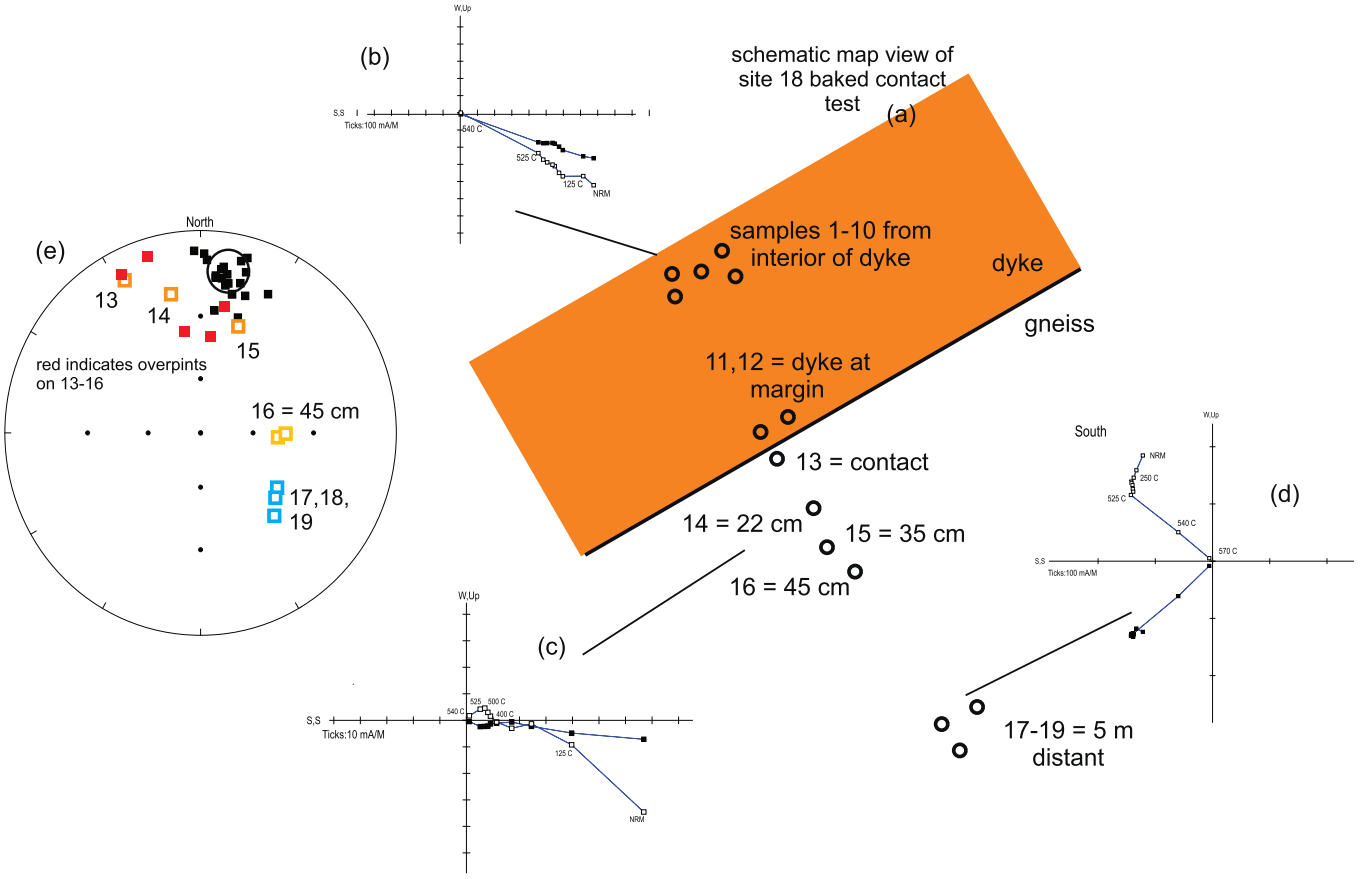


Fig. 10. Map view schematic of baked contact test at site 18 (a) along with demagnetization data from the dyke interior (b), transitional zone (c), and distant host rock (d). Summary stereoplot of direction shown in (e). This contact test is discussed in Section 3.2.3.

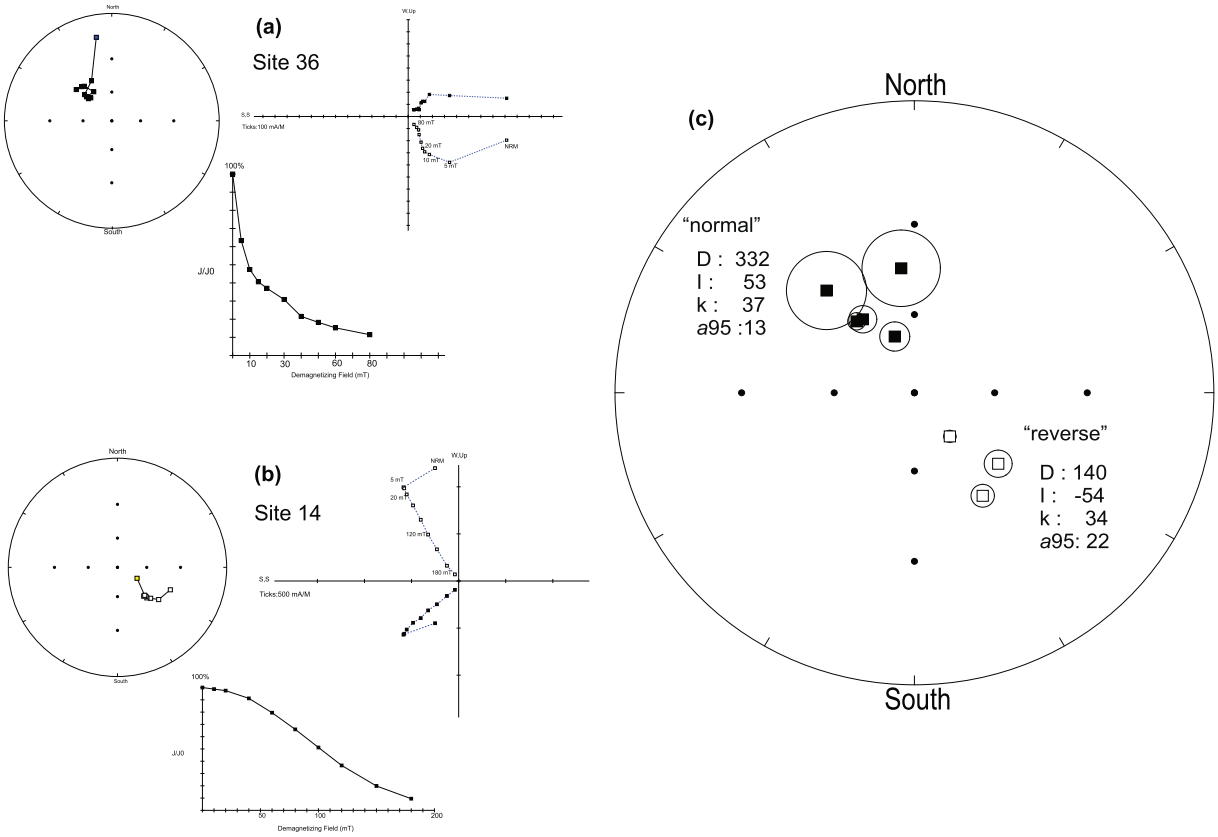


Fig. 11. Demagnetization data from bivectorial normal data (a) and near univectorial normal data (b) from a NW-SE intermediate paleomagnetic direction of unknown age. Site mean directions are shown on the stereoplot (c).

from four dykes near Ippaguda in Telegana State (NE Dharwar craton). Recently, Srivastava et al. (2016) also published a 2252 ± 2 Ma Pb-Pb baddeleyite age for a NE-SW dyke from the Singhbhum Craton, underlining a widespread nature of this apparent "swarm". French and Heaman (2010) and Kumar et al. (2012a, b) identified a ca. 2210 ± 10 Ma dyke swarm, having a relatively clear N-S trend in the northern Dharwar Craton but apparently fanning out in the south. Paleomagnetic data from Kumar et al. (2012a, b) and from Belica et al. (2014) showed the extent of this swarm across the Dharwar Craton. Our data shows a possible continuity of this swarm in the Southern Granulite Terrane, although the data are limited. Moreover, French and Heaman (2010) also defined a 2180 Ma swarm in the Dharwar Craton, based on geochronological evidence. Paleomagnetic signatures for the dykes of this age were presented in Piispa et al. (2011) and Belica et al. (2014). Our intermediate dual-polarity directions most closely resemble these data – but the paleomagnetic poles disagree; the Belica et al. (2014) pole falls at 68° N, 85° E (A95 = 18°) versus 53° N, 32° E (A95 = 11°) for our pole. As mentioned earlier in discussion of older data, relative local rotations may help explain this discrepancy, but a direct comparison awaits geochronological supporting data from dykes of this paleomagnetic direction in the SGT. Regardless, the 70 Ma stretch between three distinct episodes of dyke emplacement with good quality paleomagnetic data recovered for two of the episodes shows that for certain periods of Paleoproterozoic India movement can be precisely constrained.

Kumar et al. (2015) recognized a 2082 Ma dyke swarm emplaced around the Cuddapah Basin. Data from multiple workers and areas (Piispa et al., 2011; Belica et al., 2014; Kumar et al., 2015) provide a reasonable paleomagnetic pole for this dyke swarm.

Precise baddeleyite/zircon ages for the Pullivendla sill at 1885 ± 3.1 Ma (French et al., 2008) and NW-SE trending Bastar dykes at 1891.1 ± 0.9 Ma and 1883.0 ± 1.4 Ma (French et al., 2008). The paleomagnetic pole for the 1.88 Ga dykes in the Dharwar Craton falls at 37° N, 334° E (A95 = 7°; Belica et al., 2014). A limited study on the 1.88 Ga age NW-SE dykes from the Bastar Craton also showed a pole similar to the Dharwar Craton (31° N and 330° E; A95 = 12°; Meert et al., 2013), The extension of this dyke swarm through several cratons of the South India Block (SIB) and possibly into the Southern Granulite Terrane (this study) presents compelling evidence for their unification at this time. The potential for India movement delineation in this time period grows when a 1785 ± 14 Ma baddeleyite age from a dyke near Pebbair, north of the Cuddapah, is also considered (Demirer, 2012). Newer dolerites from the Singhbhum Craton are dated at 1765 ± 1 Ma (Shankar et al., 2014) which could mean widespread and protracted dyke emplacement in India at this time – at least from the Dharwar and Singhbhum Cratons. Our limited geochronological work in the SGT did not reveal any ~1765 Ma dykes. The mounting evidence for SIB unification into the Paleoproterozoic begs the question: is there any evidence for North India Block proximity at this time? As mentioned in the introduction, NIB-SIB unification remains an area of active research. Future combined paleomagnetic and geochronologic work focused on the North Indian Block will aid in any comparisons with the well-populated SIB "barcode".

4.3. The Northern Block of the SGT and the Dharwar Craton: a metamorphosed chip off the old block

Given that this study area is a part of the Southern Granulite Terrane, delineating a metamorphic component to the paleomagnetic data was perhaps inevitable. Tectonothermal activity in the SGT shows distinct peaks at the Archean-Proterozoic transition (Anderson et al., 2012; Collins et al., 2014), mid-Cryogenian (Ghosh et al., 2004), and early Cambrian from 540 to 525 Ma (Clark et al., 2009). The earliest major metamorphic event, at the beginning Paleoproterozoic (Anderson

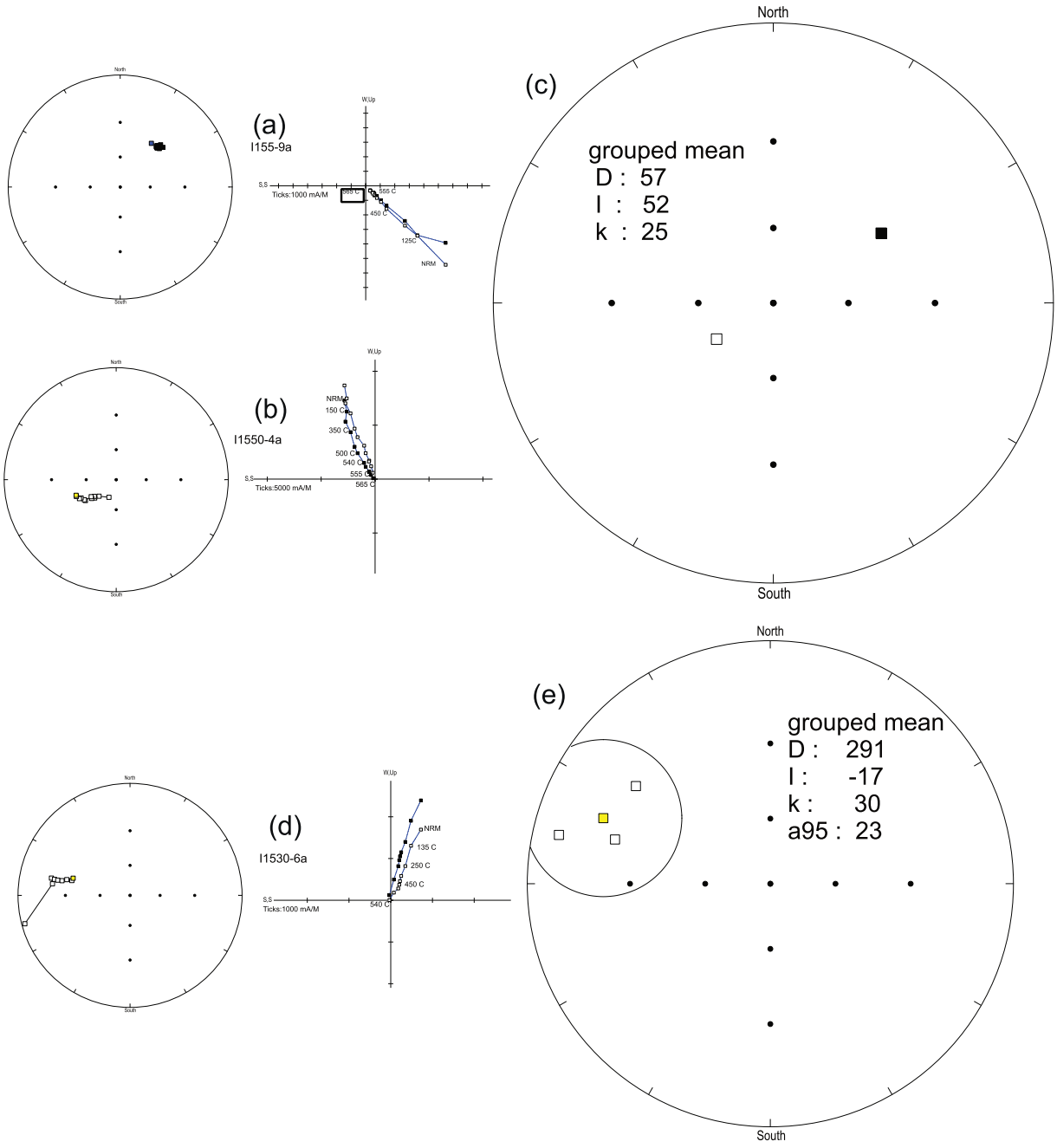


Fig. 12. Demagnetization data for dykes provisionally associated with 2210 Ma emplacement (a,b,c) and 1880 Ma emplacement (d,e). Both dykes of the former show near univectorial demagnetization paths (a, b). The latter exhibits more of an overprint (d). Site means are shown on the labelled stereoplots (c,e).

et al., 2012) pre-dates emplacement of the oldest Southern Indian dykes. Therefore, a Neoproterozoic overprint age is more likely within the Southern Granulite Terrane. We have identified several potential overprints in our study. The "B" component, yields a pole at 82° N, 267° E (A95 = 11°, Fig. 16). Combining this component with other directional data from the SGT (Halls et al., 2007; Radhakrishna and Joseph, 1996) a grand mean paleomagnetic pole is calculated at 81° N, 259° E (A95 = 6°). The "B" component pole, however, lacks direct geochronological supporting data. This widespread overprint was interpreted by both Halls et al. (2007) and Belica et al. (2014) as Ediacaran in age. An Ediacaran overprint age is consistent with other poles in the Gondwana database (Fig. 17; Table 7; Meert et al., 2001; Torsvik et al., 2012; Trindade et al., 2006; Rapalini et al., 2015; Robert et al., 2017).

During the 540–525 Ma interval, the Madurai Block collided with the Salem Block followed by orogenic collapse (Clark et al., 2009). The cryptic geochronological results from site 10 implied at least one dyke in the Tiruvannamalai area has a crystallization age at 527 ± 2.6 Ma, which implies magmatic activity at the end of the Kuunga (later called the Malagasy) Orogeny (Meert, 2003; Clark et al., 2009). An orogenic collapse would lead to extension, and dyke activity is consistent with an extensional environment. Do the paleomagnetic data from around Tiruvannamalai and elsewhere in the Southern Granulite Terrane support this scenario? Dyke T9, one of several NE-SW trending dykes cutting the large 2370 Ma dyke at sites 10, 11, and 12 has a VGP that falls at 7° N, 347°E (A95 = 5°). The East Coast Dykes (ECD) sampled by Dash et al. (2013) yielded a paleomagnetic pole at 2° S, 8° E (A95 = 6°). Additionally, two 'anomalous' overprints from the Southern Granulite Terrane yield a VGP at 7° N, 352° E (A95 = 12°). It seems likely that the ECD and at least several dykes from the Tiruvannamalai area preserve coeval paleomagnetic data at ca. 527 Ma. Paleomagnetic poles from

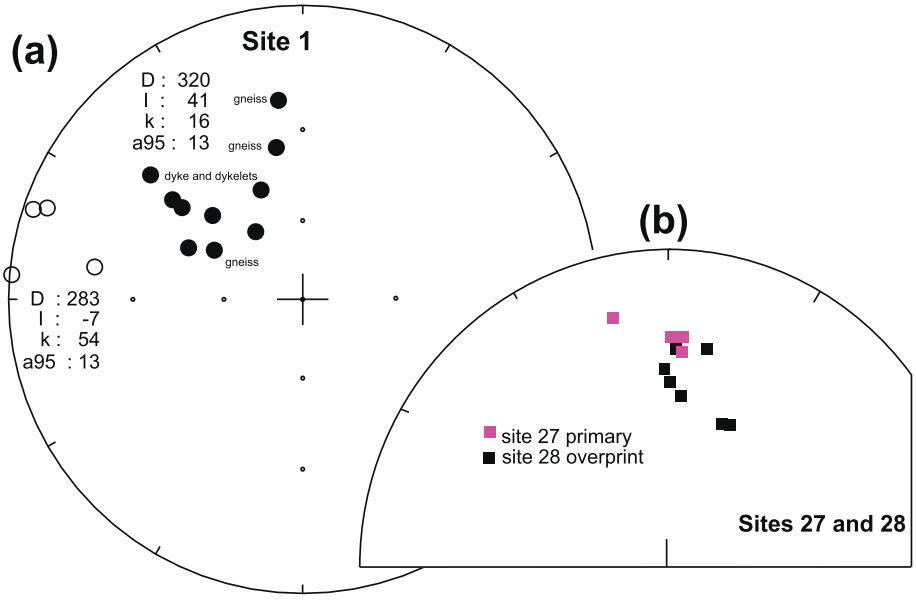


Fig. 13. Two examples of contrasting paleomagnetic data in close proximity in the Southern Dharwar craton and northern Southern Granulite Terrane. Site 1 (a) near Tirupati is at an area of cross-cutting dykes where the older dyke appears to have baked the surrounding gneiss, while a younger dyke intruded at the same outcrop, similar to the dykes at site 30a,b. Site 28 and 27 (b) are from a dyke and the associated dykelets respectively. The dykelets are completely overprinted, while the larger dyke preserves a 2.37 Ga direction after the removal of overprints.

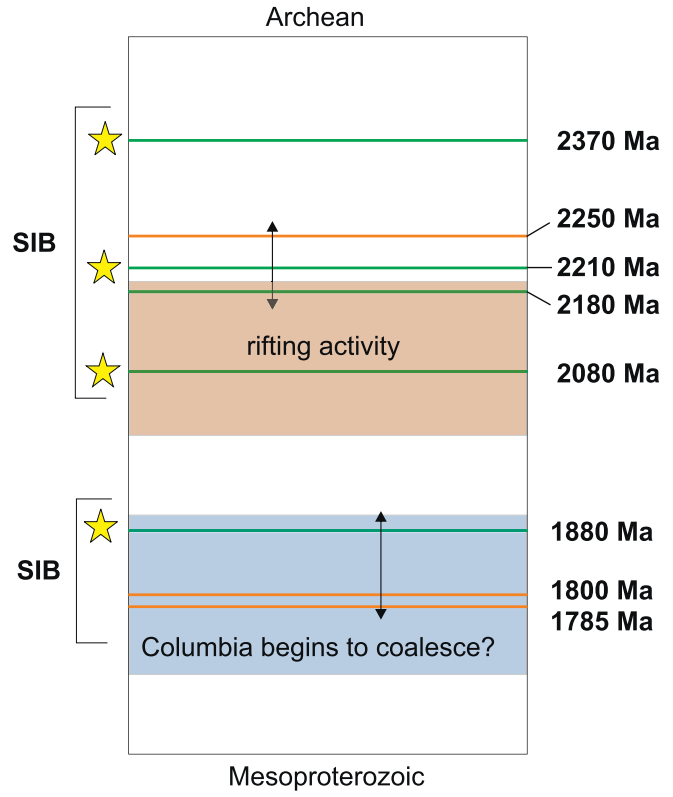


Fig. 14. 'Barcode' of igneous activity for primarily the South Indian Block. Green lines indicate both geochronological and paleomagnetic control on the igneous event. Highly studied events also have a star. Orange indicates only geochronological control. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

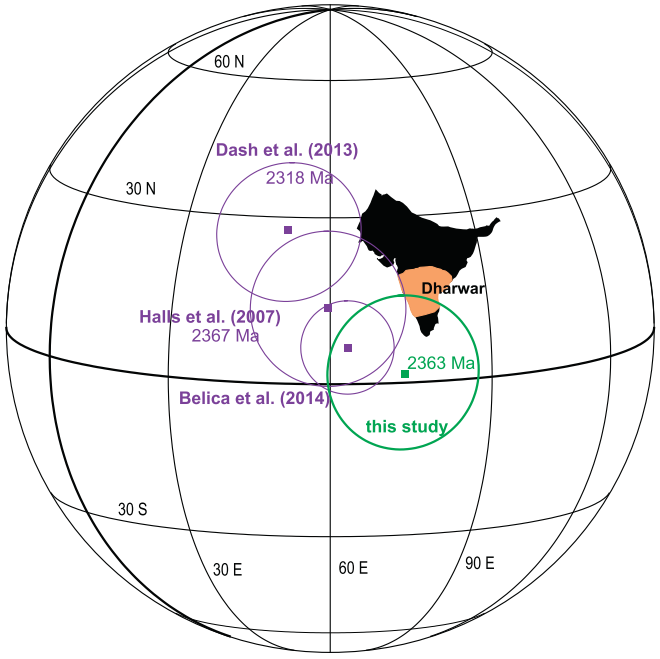


Fig. 15. Comparison of the paleomagnetic pole calculated from our primary 2363 ± 6.6 Ma paleomagnetic direction (green) with other studies from the Southern Dharwar craton (violet): Halls et al. (2007), Dash et al. (2013), and Belica et al. (2014). Our pole is lower latitude and more easterly, but overlaps with the Belica et al. (2014) pole within confidence limits. Note that purely SGT results (Dash et al., 2013; this study) may be rotated with respect to results from farther northward in the Dharwar Craton. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

stratoid granites (7° S, 353° E; A95 = 14°; 521.4 ± 11.9 Ma; Meert et al., 2003) and the Carion granite (7° S, 001° E; A95 = 15°; 508.5 ± 11.5 Ma; Meert et al., 2001) in Madagascar show excellent agreement with the Gondwana APWP when rotated into an African reference frame (Fig. 17) and are roughly coeval in time with our proposed magmatism in the SGT. However, when rotated into an African reference frame, it is clear that our proposed 527 Ma Indian pole does not agree with the Gondwana APWP (Fig. 17). Since the ECD and Tiruvannamalai data are all from the same block of the SGT east of the Mettur shear zone (Fig. 2), our earlier discussion of possible clockwise rotation of this block may hold for these data as well. However, as mentioned earlier, late Ediacaran paleomagnetic data from the Marwar Supergroup in NW India is roughly coeval with the SGT data (Davis et al., 2014; Fig. 17), which makes significant rotation at this time less likely. We note an interesting resemblance between the angular shift between our Ediacaran "B" component pole with the proposed early Cambrian Indian paleomagnetic results and the large-scale apparent polar wander curve shifts around the Ediacaran-Cambrian

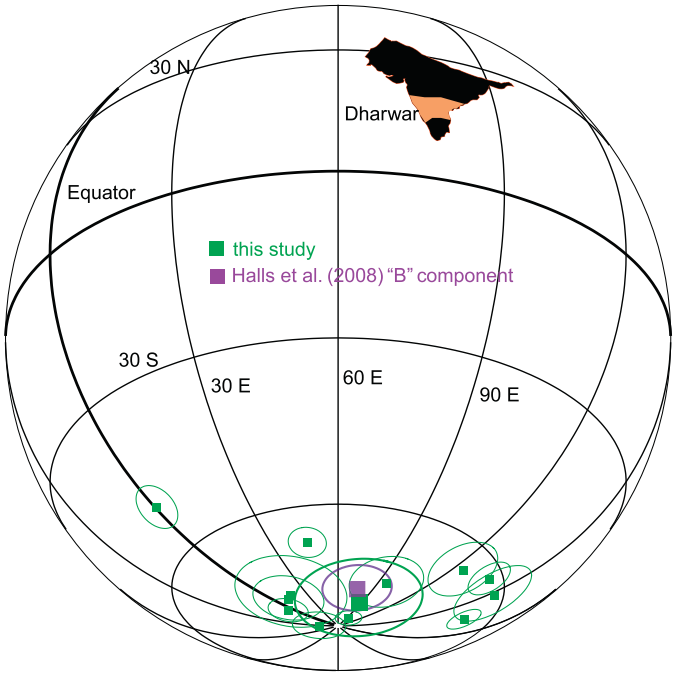


Fig. 16. Poles calculated from the shallow paleomagnetic component seen in 12 dykes of our study. Mean pole is compared to the 'B' component of Halls et al. (2007).

discussed at length in Robert et al. (2017). However, the geochronological constraints on our data provide a longer time window for this possible shift (mid-Ediacaran – 527 Ma).

Early attempts at defining geologic provinces in India (Fermor, 1936; Drury et al., 1984) considered the SGT to extend to the roughly-defined orthopyroxene-in isograd, or the Fermor Line. General structural continuity between the Northern Block and the greater Dharwar craton provides evidence of their contiguity (Meissner et al., 2002). Crustal thickness and seismic studies (Rao and Prasad, 2006) suggest that the Northern Block collided with the Dharwar Craton in the late

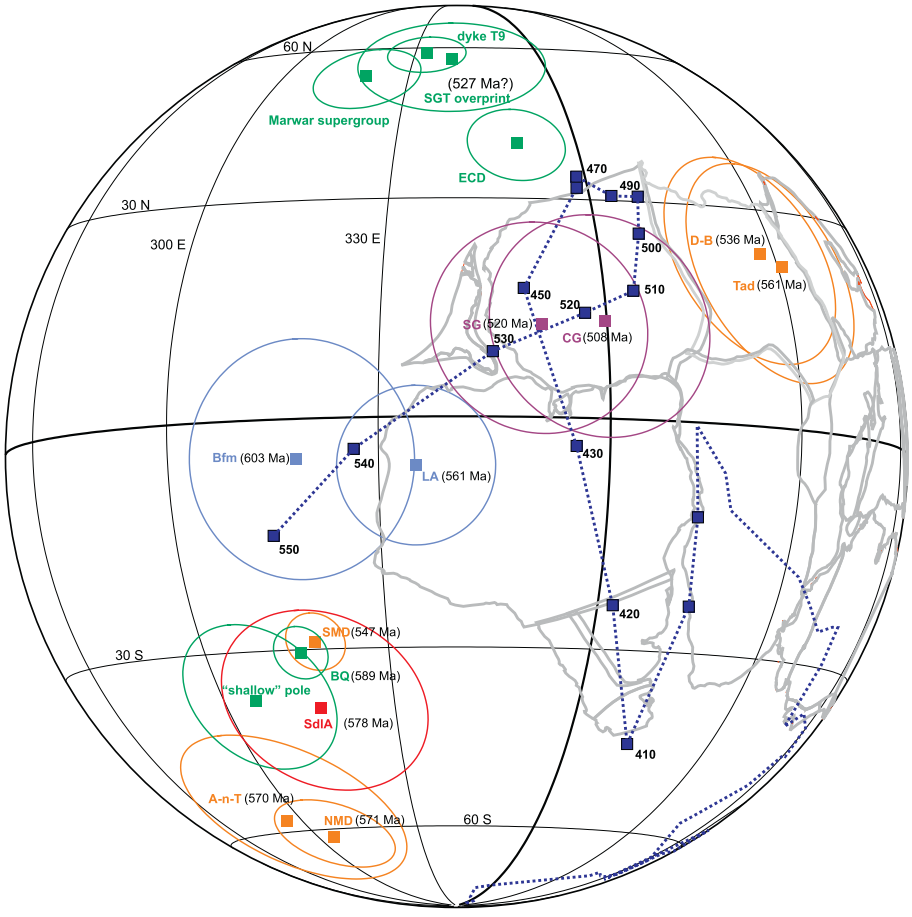


Fig. 17. Ediacaran – Early Cambrian poles and VGPs from Southern India compared to paleomagnetic data of the same time interval from other Gondwana blocks (see Table 7 for detailed information and rotations parameters). Indian poles are shown in green, Australian in baby blue, African in orange, and Madagascar in violet. The running mean Gondwana APWP of Torsvik et al. (2012) is shown by the dashed blue line, with age squares until 390 Ma provided. Outlines of Gondwana blocks shown in light gray. Note that TD and the overprint direction represent VGPs, and not paleomagnetic poles proper. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Archean (Meert et al., 2010, 2011). The 2538 ± 6 Ma and 2529 ± 7 Ma SHRIMP Pb-Pb ages of zircons from charnockitic rocks (Clark et al., 2009) with younger ages obtained from the zircon rims (ca. 2480 Ma) neatly bracket the general stabilization age of the Dharwar Craton. This either shows that the Northern Block was contiguous with the Dharwar Craton earlier and stabilized at the same time, or that their amalgamation in the Late Archean was an important part of the stabilization process. Regardless of which interpretation is correct, we concur with Dash et al. (2013) in viewing the early Proterozoic dyke swarms present in both the Northern Block of the SGT and the Dharwar as unequivocal evidence supporting their contiguity at the time of emplacement.

The well-confirmed high-latitude position for the Dharwar Craton (now shown to include the SGT down to the PCSZ – e.g. the Northern Block) allows for the testing of Paleoproterozoic cratonic reconstructions. Halls et al. (2007) tentatively reconstructed the Yilgarn and Dharwar cratons based on paleomagnetic and geochronologic data from the Widgiemooltha swarm and the Bangalore swarm, respectively. Both the swarms are tholeiitic in composition and are indistinguishable in a Ti-Zr-Y discrimination plot which may indicate a genetic link between them given the identical concentrations of the three elements (Halls et al., 2007). The mean 2.37 Ga Dharwar pole from this study is at 2° N and 73° E (A95 = 14°) and the Yilgarn pole from the Widgiemooltha dykes is at 10° N, 339° E (A95 = 8°; Evans, 1968; Smirnov et al., 2013). Reconstructing the Dharwar and Yilgarn according to these poles (given hemispheric ambiguity and longitudinal uncertainty) allows the two cratons to be placed together. Although an argument was forwarded to correlate these two intrusive events by Halls et al. (2007), we believe the geochronological data are problematic for that correlation. Age data from the Bangalore swarm (Halls et al., 2007; Kumar et al., 2012b; this study) points to rapid emplacement, with all ages falling between 2360 and 2370 Ma. The Widgiemooltha swarm, on the other hand, was emplaced around 2410–2418 Ma (Nemchin and Pidgeon, 1998; French et al., 2002). A study on the 2401 ± 1 Ma Erayinia dykes from the Yilgarn Craton produced a paleomagnetic pole at 23° N, 331° E (A95 = 11°; Pisarevsky et al., 2015). Pisarevsky et al. (2015) calculated an angular velocity of ~1°/Ma for the Yilgarn craton. Therefore, the 2.40–2.41 Ga correlations shown in Pisarevsky et al. (2015) are not necessarily applicable to the Dharwar Craton since the dykes there are ca. 30 Myr older than the dykes of the Yilgarn.

5. Conclusions

1. Dykes in the northern part of the Southern Granulite Terrane reflect multiple intrusive pulses ranging in age from the earliest Paleoproterozoic into the Cambrian. Although this area was subjected to a higher degree of metamorphism than other areas in India, it still contains a remarkably complete Proterozoic paleomagnetic record for Southern India.
2. Our study provides ample evidence to indicate juxtaposition of SGT with Dharwar Craton from at least 2370 Ma, also mentioned earlier by Dash et al. (2013). Our U-Pb zircon age of 2363 ± 6.6 Ma precisely ties steep paleomagnetic directions from *E*-W and NW-SE mafic dykes in the SGT to steep paleomagnetic directions from E-W mafic dykes in the Dharwar Craton. Furthermore, we report more limited paleomagnetic results on dykes from other phases of magmatic activity in Proterozoic India, namely, at 2210 Ma, 2180 Ma, and 1880 Ma.
3. The Dharwar craton and its extension into the northern SGT was positioned at high latitudes in the early Paleoproterozoic (ca. 2370 Ma).
4. We have placed a tentative date of 527 ± 2.6 Ma on the East Coast Dykes (Dash et al., 2013). The geographic extent of these dykes is still largely unconstrained in the SGT, but occurs at least as far west as Tiruvannamalai. Direct dating of the East Coast dykes and other NE-SW dykes in Tiruvannamalai would be a simple and effective test of this correlation.

Acknowledgments

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